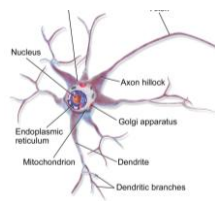


计算机视觉——深度学习简介

2026年春季
桑农 王岳环



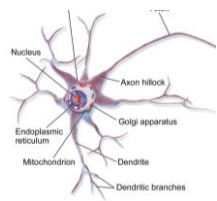
内容

深度网络介绍

- 目标识别的传统方法与深度学习方法
- 神经学习
- 卷积深度网络

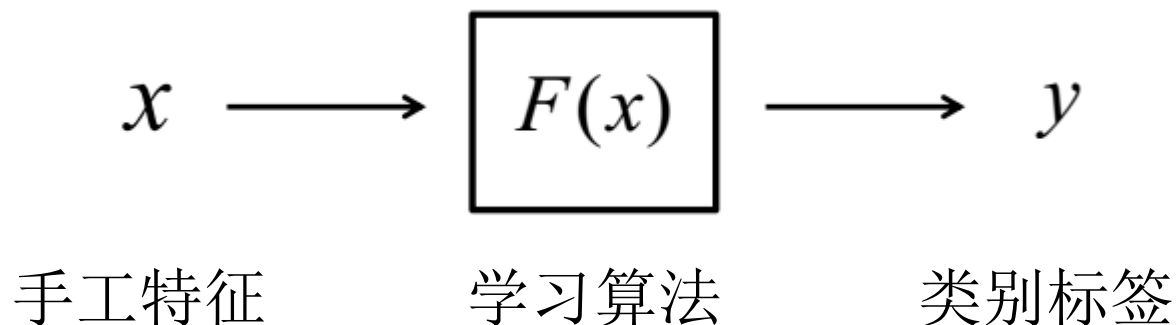
计算机视觉中的深度学习

- 计算机视觉中的深度网络
- 计算机视觉中的深度网络框架（不同任务）

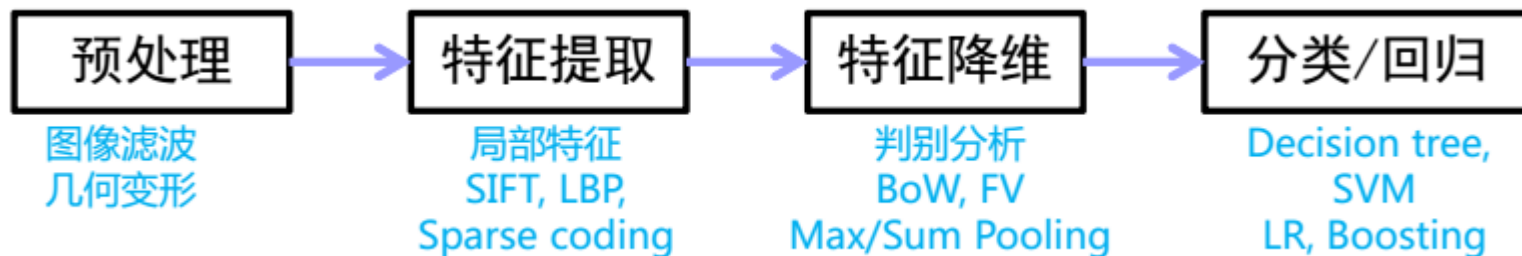


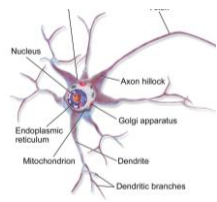
目标识别的传统方法

- 目标识别传统方法框架



- 分段处理策略





目标识别的传统方法

- 手工特征提取方法

- Haar

- HoG

- 灰度共生矩

- Shape context

- 不变距

- 颜色直方图

纹理特征

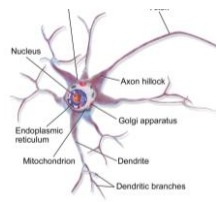
形状特征

颜色特征

知识驱动

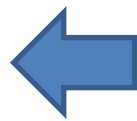
依赖经验





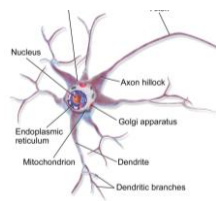
目标识别的传统方法

- 分类算法学习
 - SVM
 - Adaboost
 - KNN
 - 随机森林
 - 决策树
 - 神经网络（浅层）



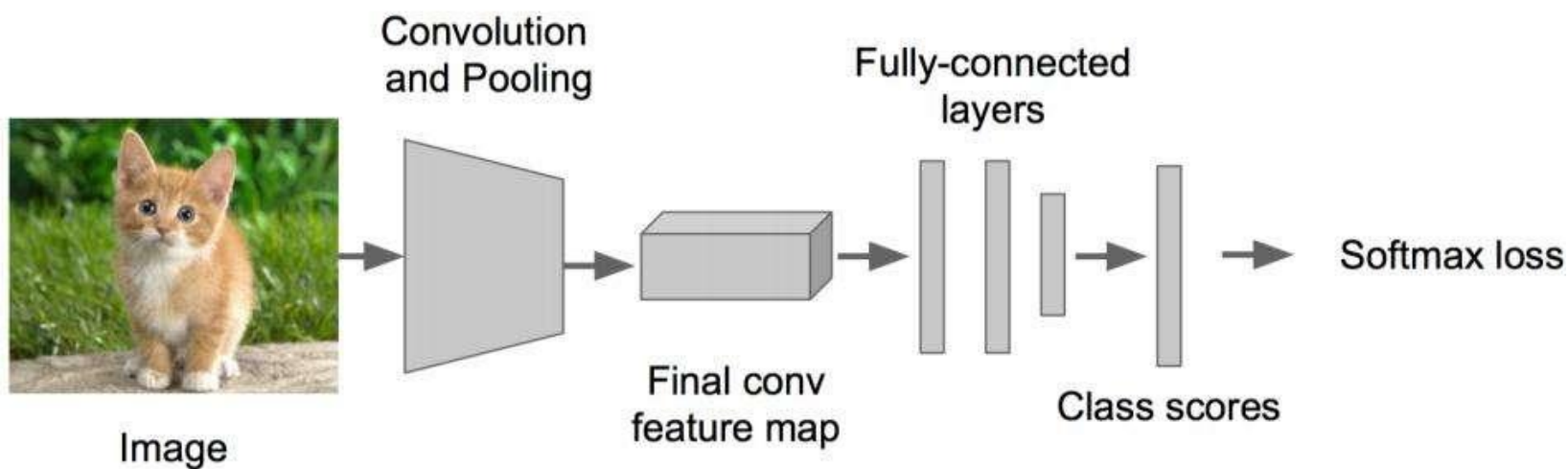
设计复杂

依赖经验



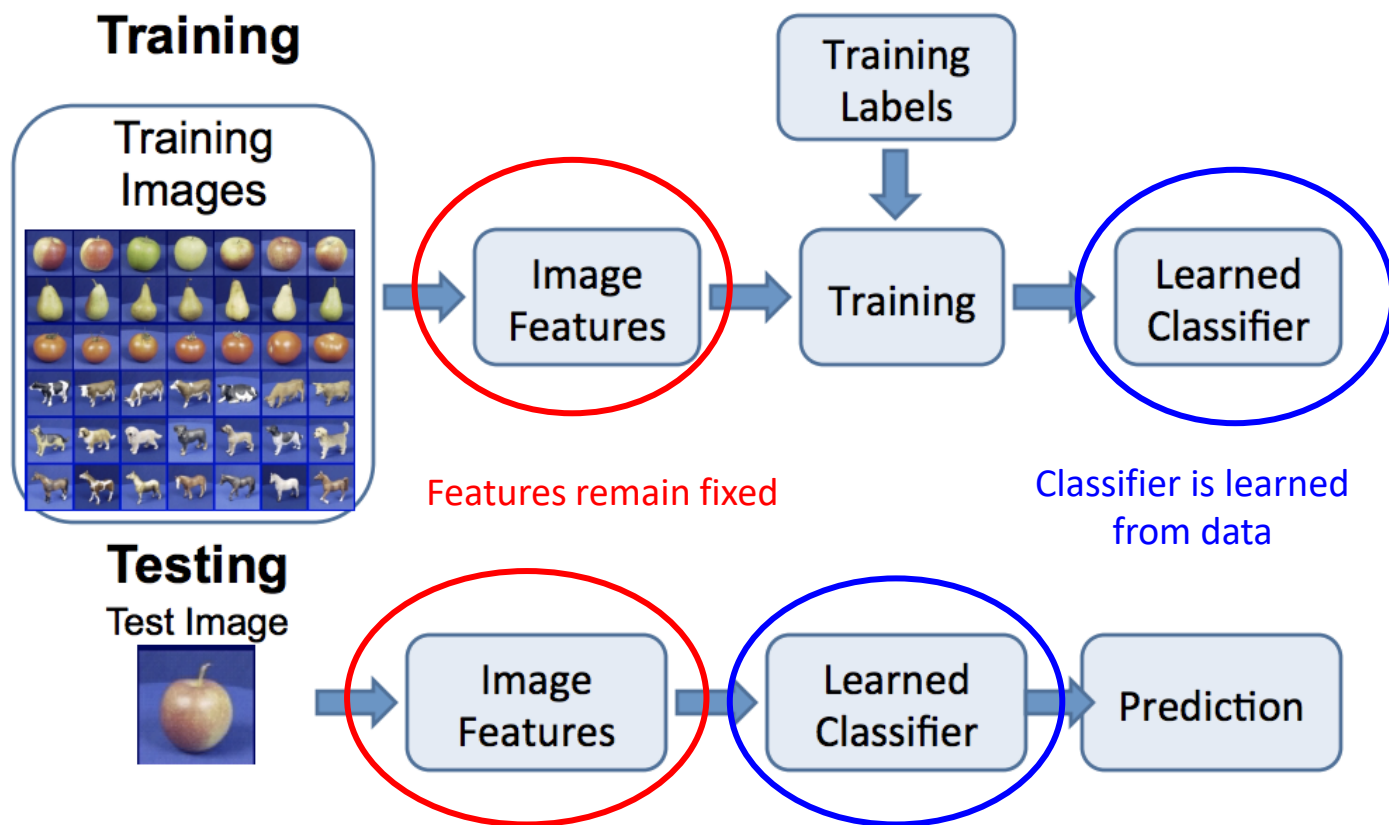
目标识别的深度学习方法

- 目标识别的深度学习框架



- 端到端的处理策略
 - 特征与分类算法协同学习

传统目标识别框架



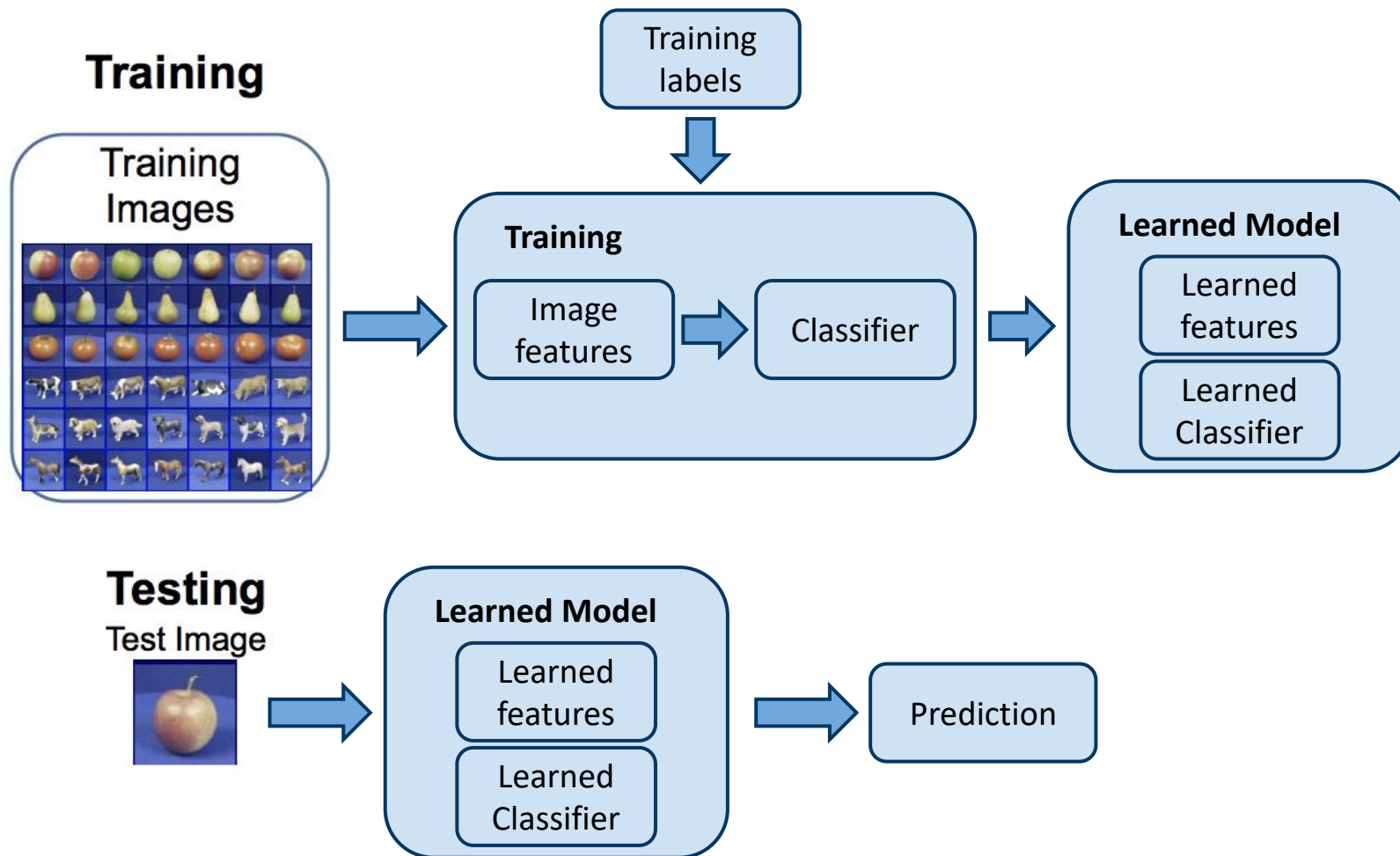
难点:
如何设计特征?
需要为不同的任务设计不同的特征!

解决方案:
同时学习特征和分类器!

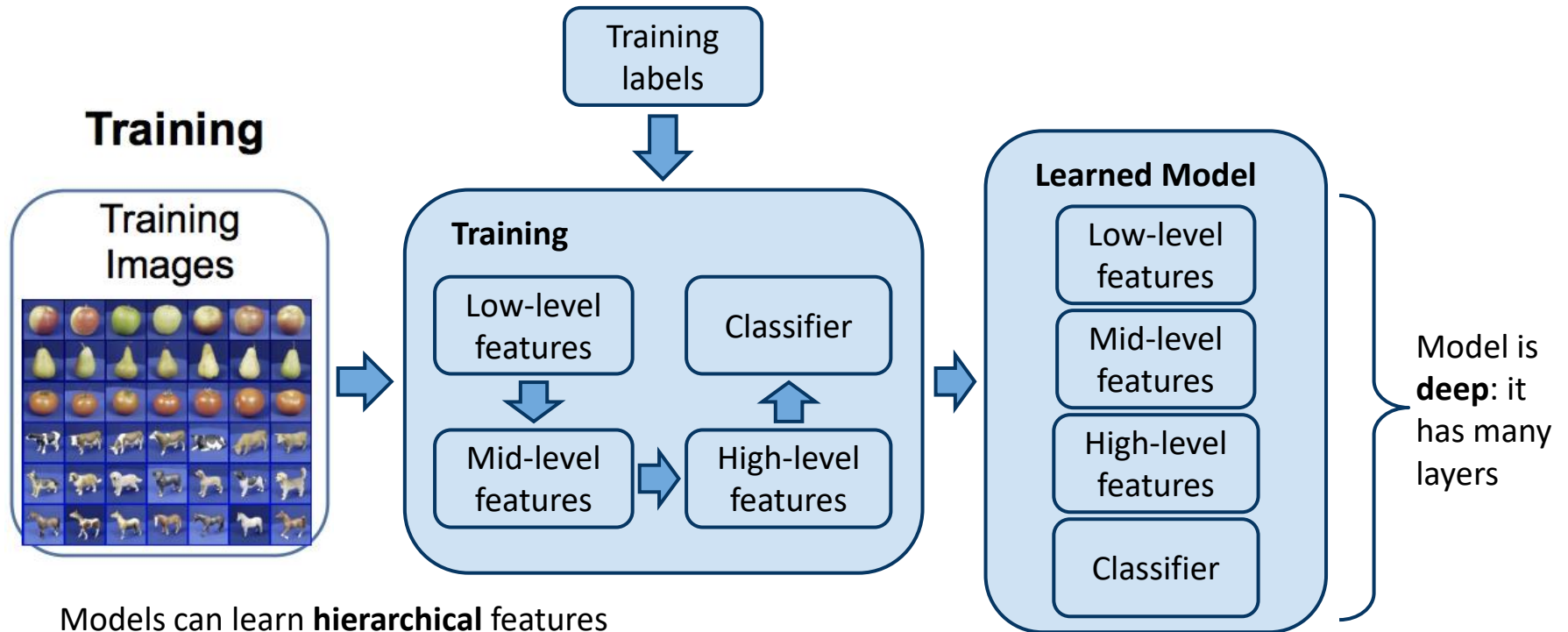
Figure credit: D. Hoiem, L. Lazebnik

6 Dec 2016

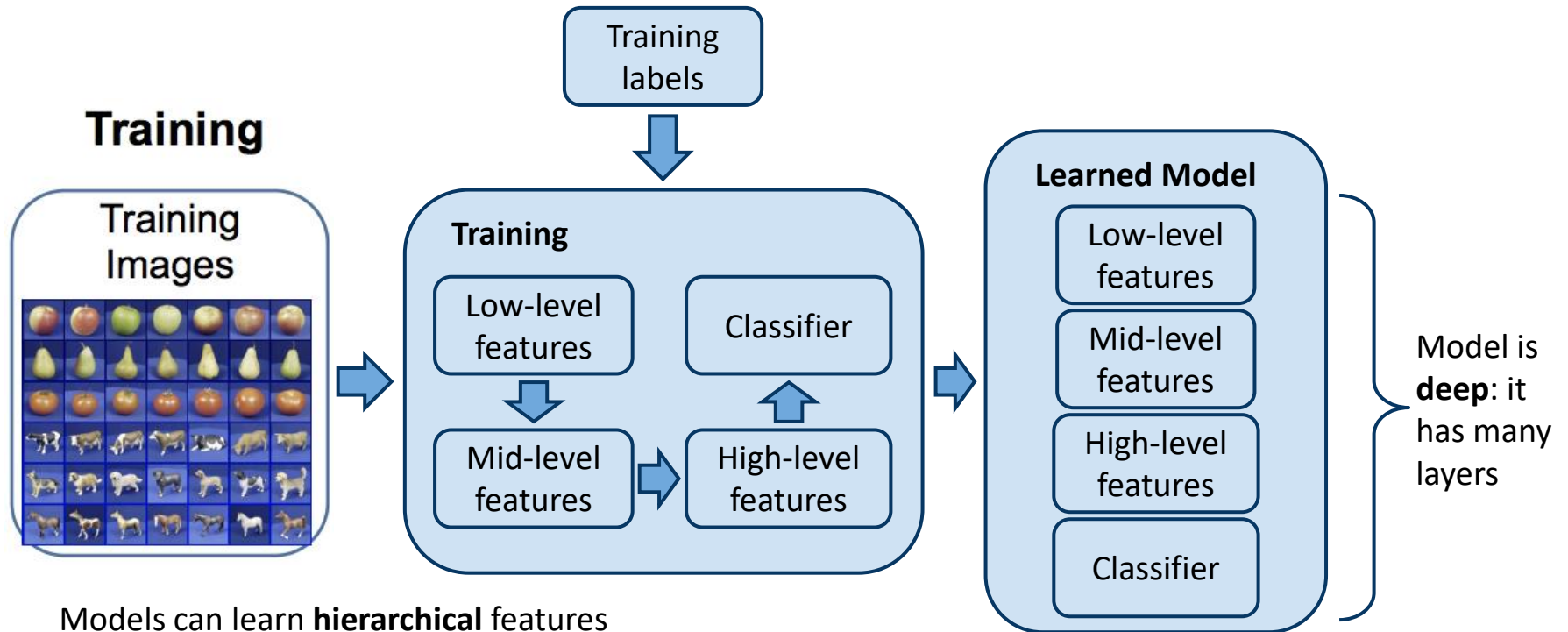
特征学习



深度学习



深度学习



Models can learn **hierarchical** features

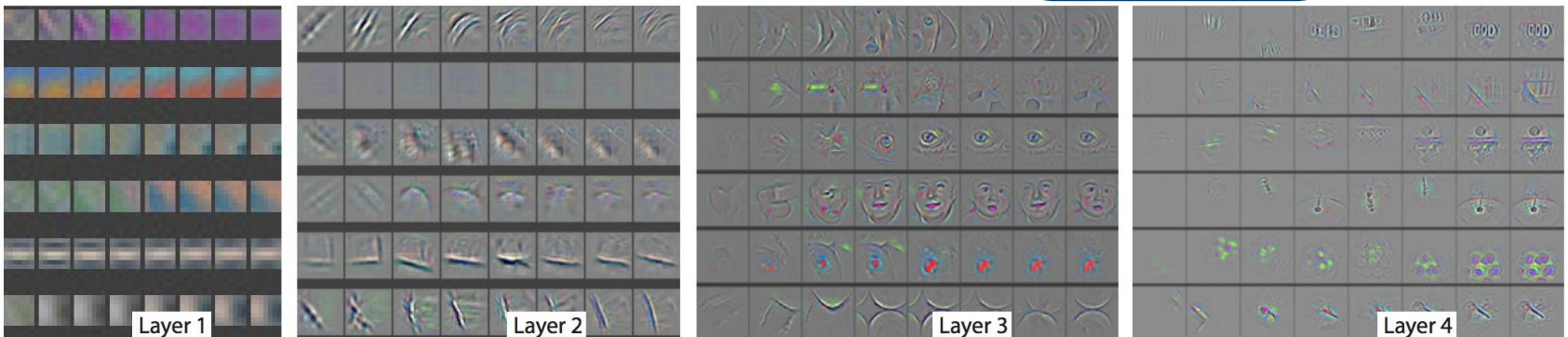
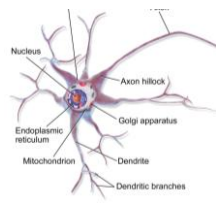


Figure credit: Zeiler and Fergus, "Visualizing and Understanding Convolutional Networks", ECCV 2014



目标识别的深度学习方法

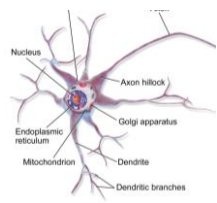
- 特征与分类算法协同学习
 - 深度学习的高性能主要依赖强大特征学习能力
 - 层次化表达
 - 高度非线性
 - 数据驱动
 - 非全局最优



可解释性差

依赖数据

参数难调

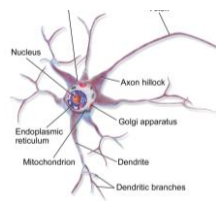


目标识别:传统方法 VS 深度学习

传统方法	深度学习
依赖经验	很强的拟合能力
设计复杂	计算复杂
小样本泛化能力强	依赖数据（解空间大）
理论完善	缺乏理论，难解释，容易陷入局部最优

性能更强

随着数据增加性能仍在上升



深度学习步骤

步骤 1:
定义一组
函数

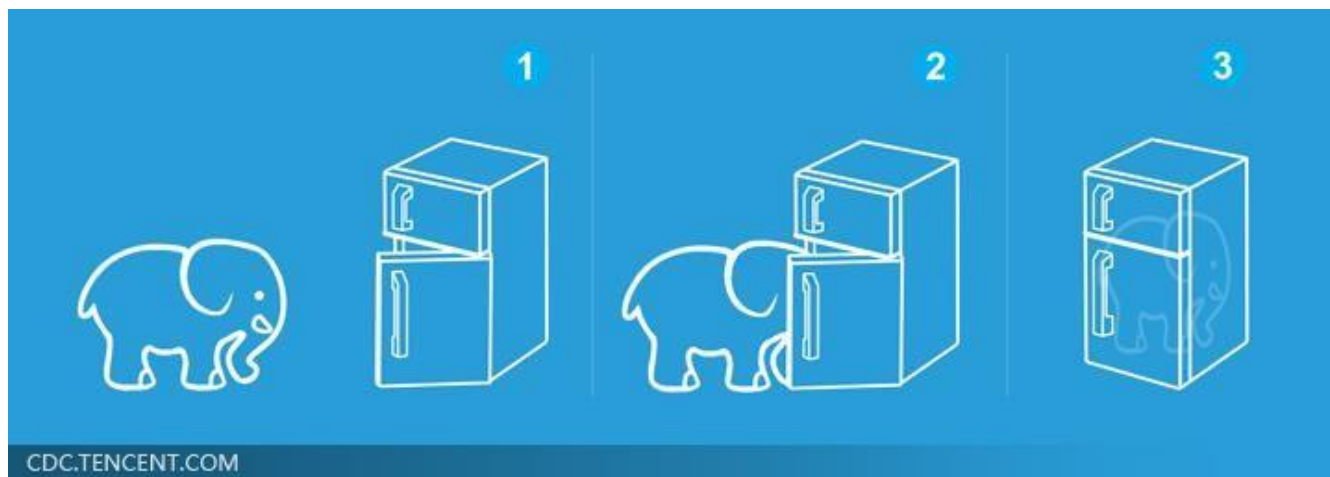


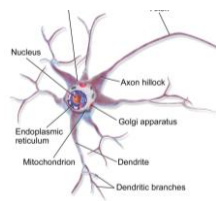
步骤 2:
模型和数
据的拟合



步骤 3:
选择最优
函数

深度学习是如此简单.....





更多任务

- 语音识别

$$f(\text{audio waveform}) = \text{"How are you"}$$

- 图片识别

$$f(\text{cat image}) = \text{"Cat"}$$

- 下围棋

$$f(\text{go board state}) = \text{"5-5"} \\ (\text{下一步棋})$$

- 对话系统

$$f(\text{"Hi"} \\ (\text{你所说的内容})) = \text{"Hello"} \\ (\text{系统的回答})$$

- 自动驾驶

$$f(\text{road scene image}) = \text{方向盘角度}$$

监督学习

Step 1:

Define Model

$$\hat{y} = f(x, w)$$

Predicted output (image label) Model structure Input data (Image pixels) Model weights

Step 2: Collect data

$$\{(x_i, y_i)\}_{i=1}^N$$

Training input True output

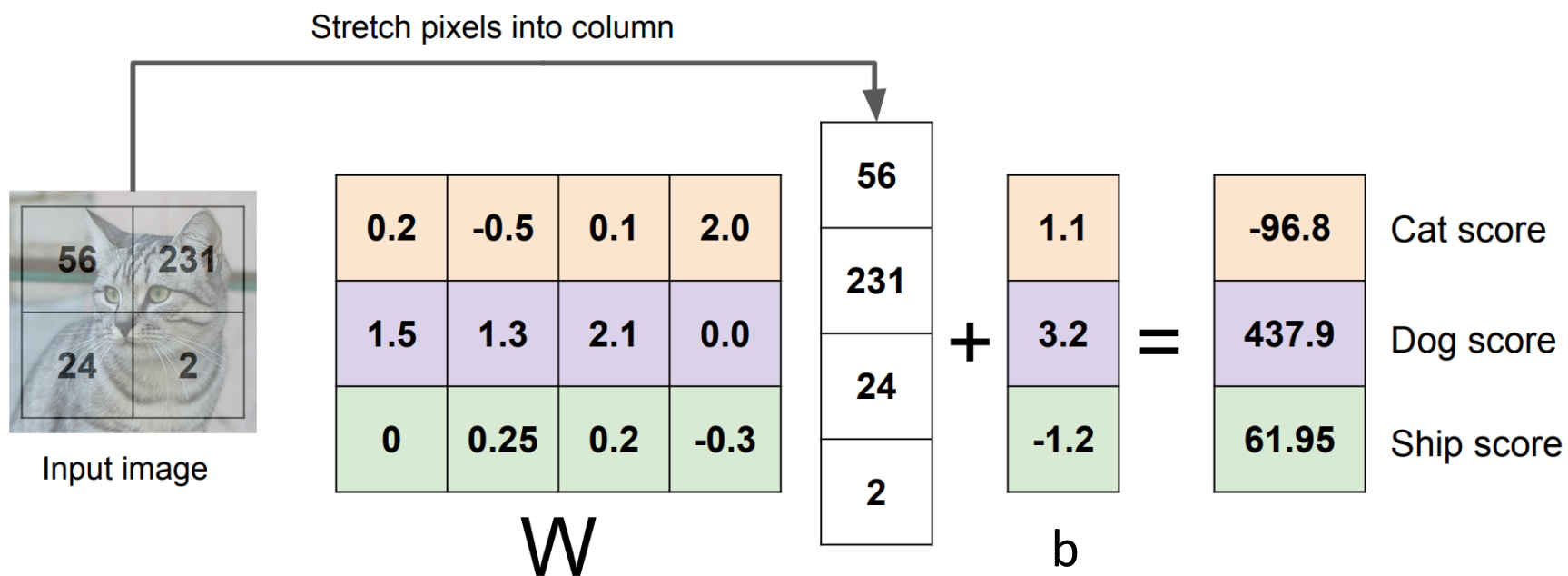
Step 3: Learn the model

$$w^* = \arg \min_w \frac{1}{N} \sum_{i=1}^N \ell(\overbrace{f(x_i, w)}^{\text{Predicted output}}, y_i) + R(w)$$

Learned weights Minimize average loss over training set Loss function: Measures "badness" of prediction Regularizer: Penalizes complex models

监督学习

Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



监督学习：神经网络

$$x_i \in \mathbb{R}^{D_{in}} \quad y_i \in \mathbb{R}^{D_{out}}$$

Input and output are vectors

$$\ell(\hat{y}, y) = \frac{1}{2} \|\hat{y} - y\|_2^2$$

Loss is Euclidean distance

Linear Regression

$$f(x, W) = Wx$$

$$W \in \mathbb{R}^{D_{out} \times D_{in}}$$

Model is just a matrix multiply

New Model

$$f(x, W_1, W_2) = W_2 W_1 x$$

$$W_1 \in \mathbb{R}^{H \times D_{in}}$$

$$W_2 \in \mathbb{R}^{D_{out} \times H}$$

Model is **two** matrix multiplies

Question: Is the new model “more powerful” than Linear Regression? Can it represent any functions that Linear Regression cannot?

Answer: NO! We can write $W = W_2 W_1$
And recover Linear Regression

监督学习：神经网络

$$x_i \in \mathbb{R}^{D_{in}} \quad y_i \in \mathbb{R}^{D_{out}}$$

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Model is just a matrix multiply

Neural Network

~~$$f(x, W_1, W_2) = W_2 W_1 x$$~~

$$f(x, W_1, W_2) = W_2 \sigma(W_1 x)$$

$$W_1 \in \mathbb{R}^{H \times D_{in}}$$

$$W_2 \in \mathbb{R}^{D_{out} \times H}$$

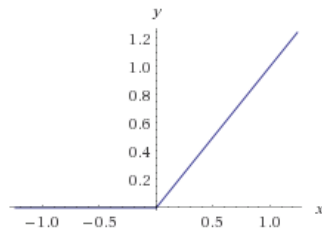
Model is **two** matrix multiplies, with an **elementwise nonlinearity**

$$\sigma : \mathbb{R}^H \rightarrow \mathbb{R}^H$$

Neural Network is more powerful than Linear Regression!

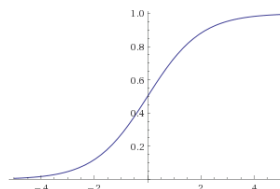
Common nonlinearities:

$$\sigma(x) = \max(0, x)$$



Rectified Linear (ReLU)

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Logistic Sigmoid

多层神经网络

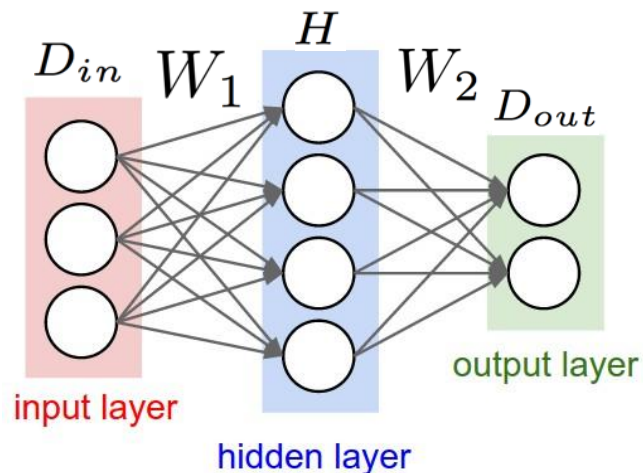
$$x_i \in \mathbb{R}^{D_{in}} \quad y_i \in \mathbb{R}^{D_{out}}$$

Two Layer network

$$f(x, W_1, W_2) = W_2 \sigma(W_1 x)$$

$$W_1 \in \mathbb{R}^{H \times D_{in}}$$

$$W_2 \in \mathbb{R}^{D_{out} \times H}$$



多层神经网络

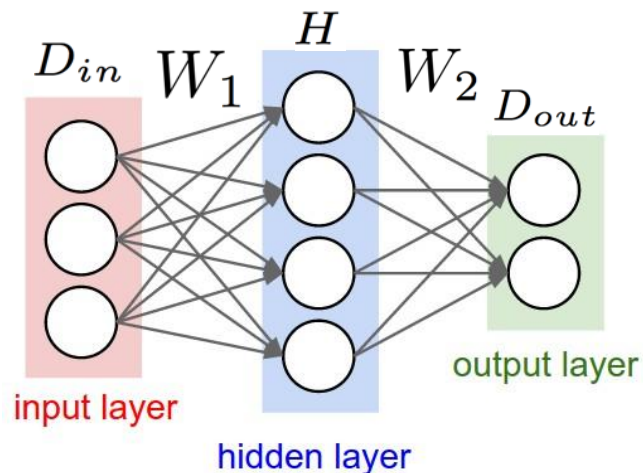
$$x_i \in \mathbb{R}^{D_{in}} \quad y_i \in \mathbb{R}^{D_{out}}$$

Two Layer network

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$$W_1 \in \mathbb{R}^{H \times D_{in}}$$

$$W_2 \in \mathbb{R}^{D_{out} \times H}$$



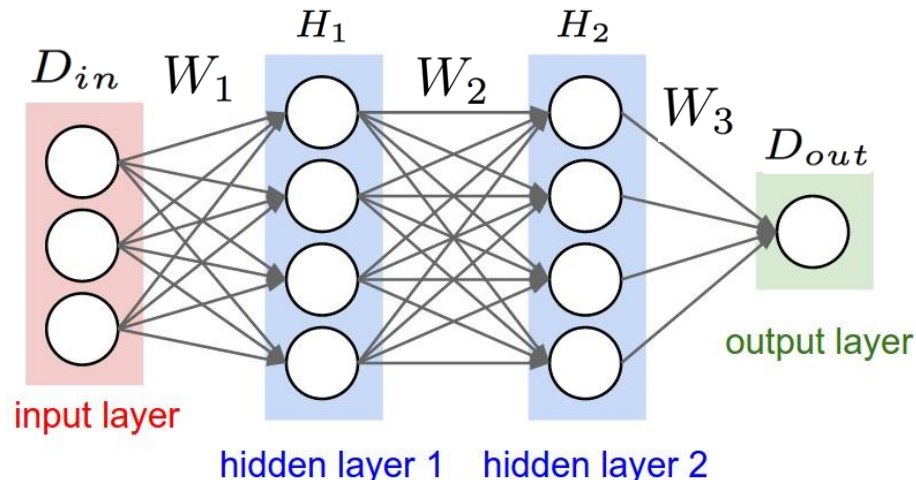
Three Layer network

$$f(x, W_1, W_2, W_3) = W_3 \sigma(W_2(\sigma(W_1 x)))$$

$$W_1 \in \mathbb{R}^{H_1 \times D_{in}}$$

$$W_2 \in \mathbb{R}^{H_2 \times H_1}$$

$$W_3 \in \mathbb{R}^{D_{out} \times H_2}$$

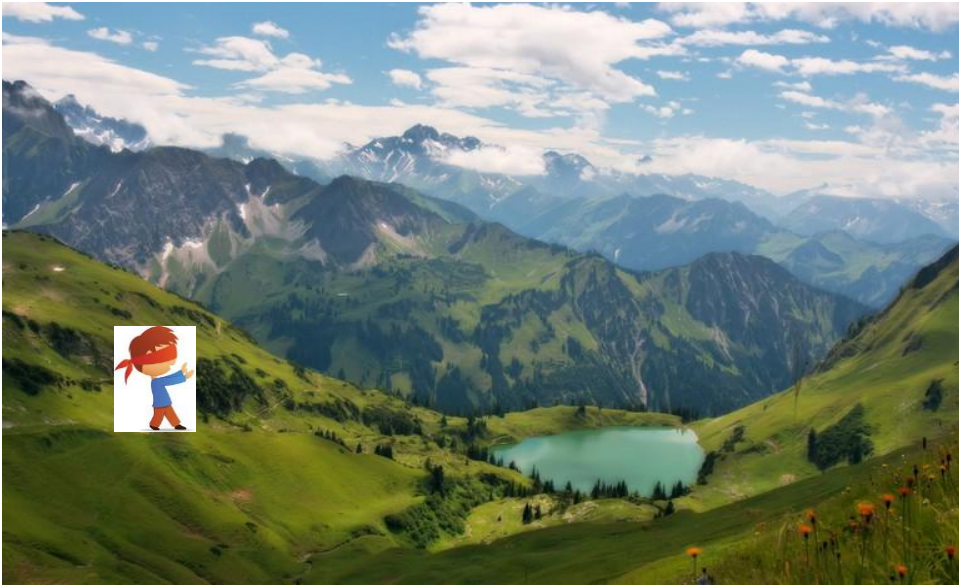


如何获得最好的 w^*

$$\begin{aligned} w^* &= \arg \min_w \frac{1}{N} \sum_{i=1}^N \ell(f(x_i, w), y_i) + R(w) \\ &= \arg \min_w g(w) \end{aligned}$$

如何获得最好的 w^*

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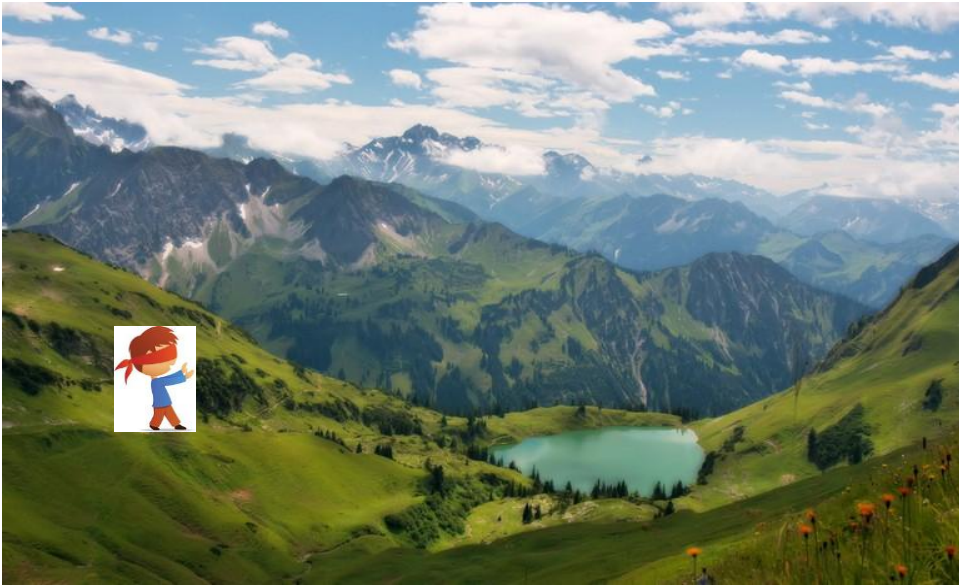


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Idea #1: Closed Form Solution

Works for some models (linear regression) but in general very hard (even for one-layer net)



如何获得最好的 w^*

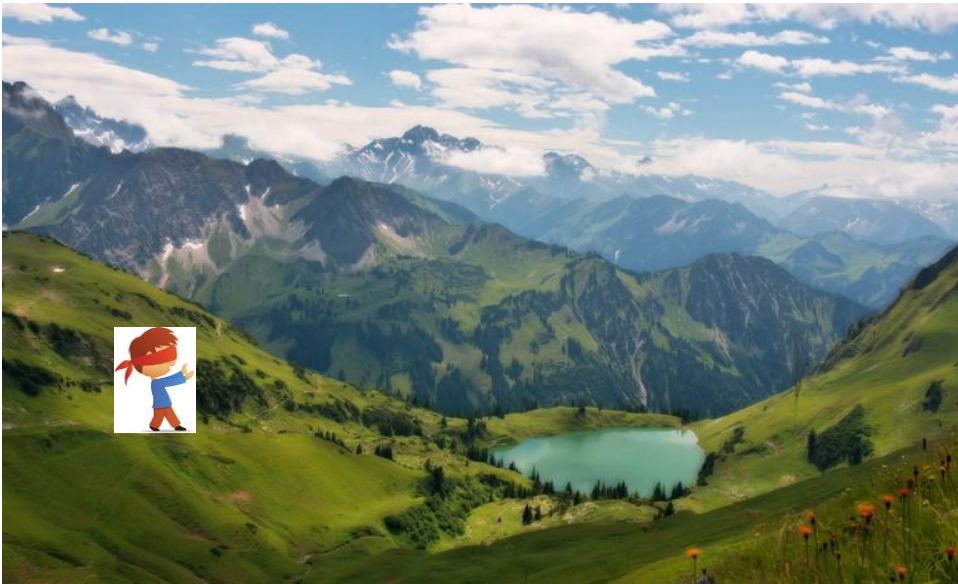
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Idea #1: Closed Form Solution

Works for some models (linear regression) but in general very hard (even for one-layer net)

Idea #2: Random Search

Try a bunch of random values for w ,
keep the best one
Extremely inefficient in high dimensions



如何获得最好的 w^*

$$w^* = \arg \min_w \frac{1}{N} \sum_{i=1}^N \ell(f(x_i, w), y_i) + R(w)$$
$$= \arg \min_w g(w)$$

Idea #1: Closed Form Solution

Works for some models (linear regression) but in general very hard (even for one-layer net)

Idea #2: Random Search

Try a bunch of random values for w , keep the best one
Extremely inefficient in high dimensions

Idea #3: Go downhill

Start somewhere random, take tiny steps downhill and repeat
Good idea!



梯度下降

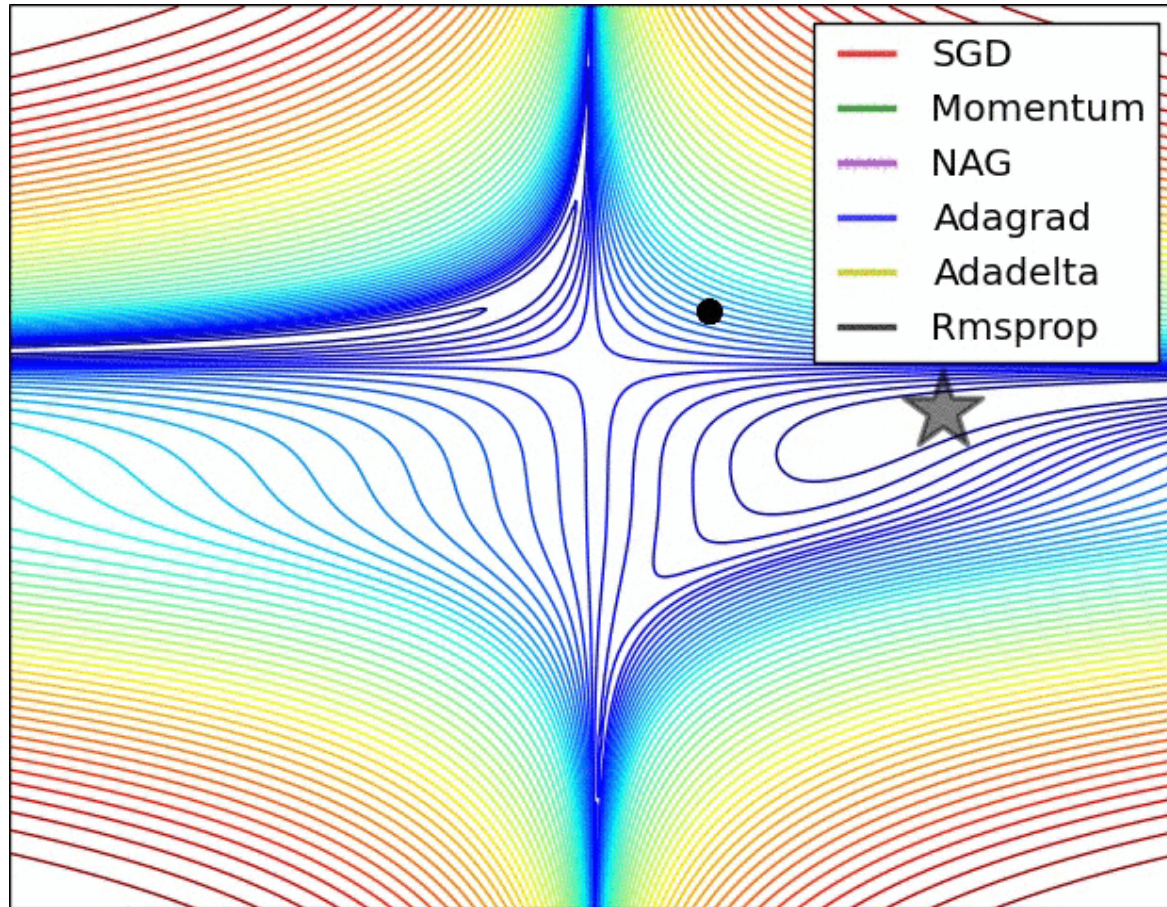
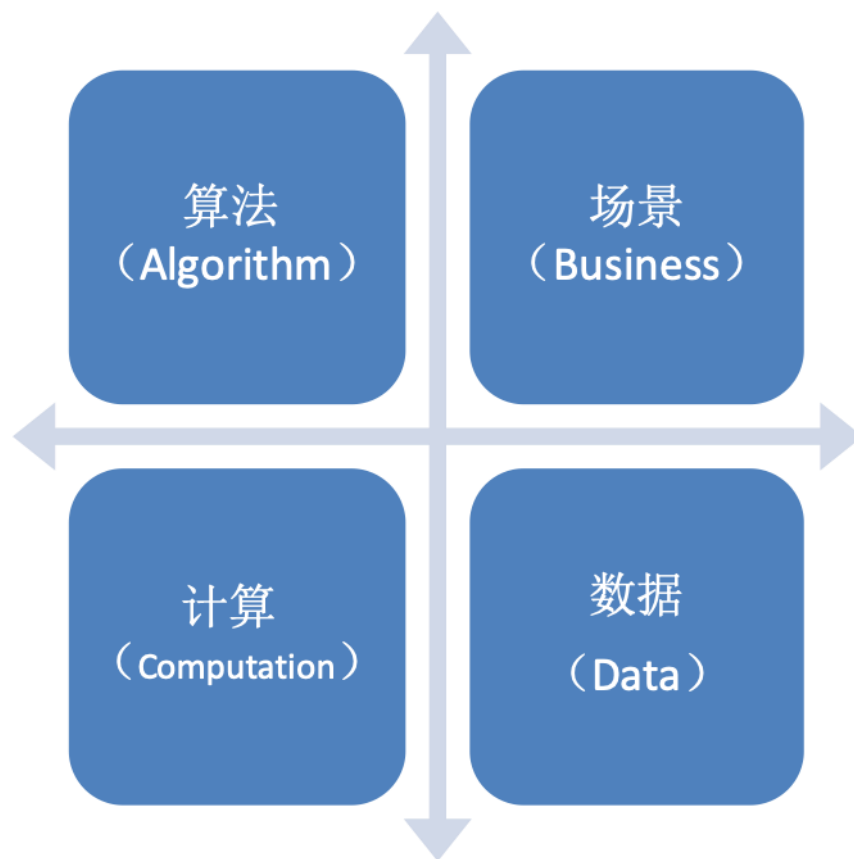


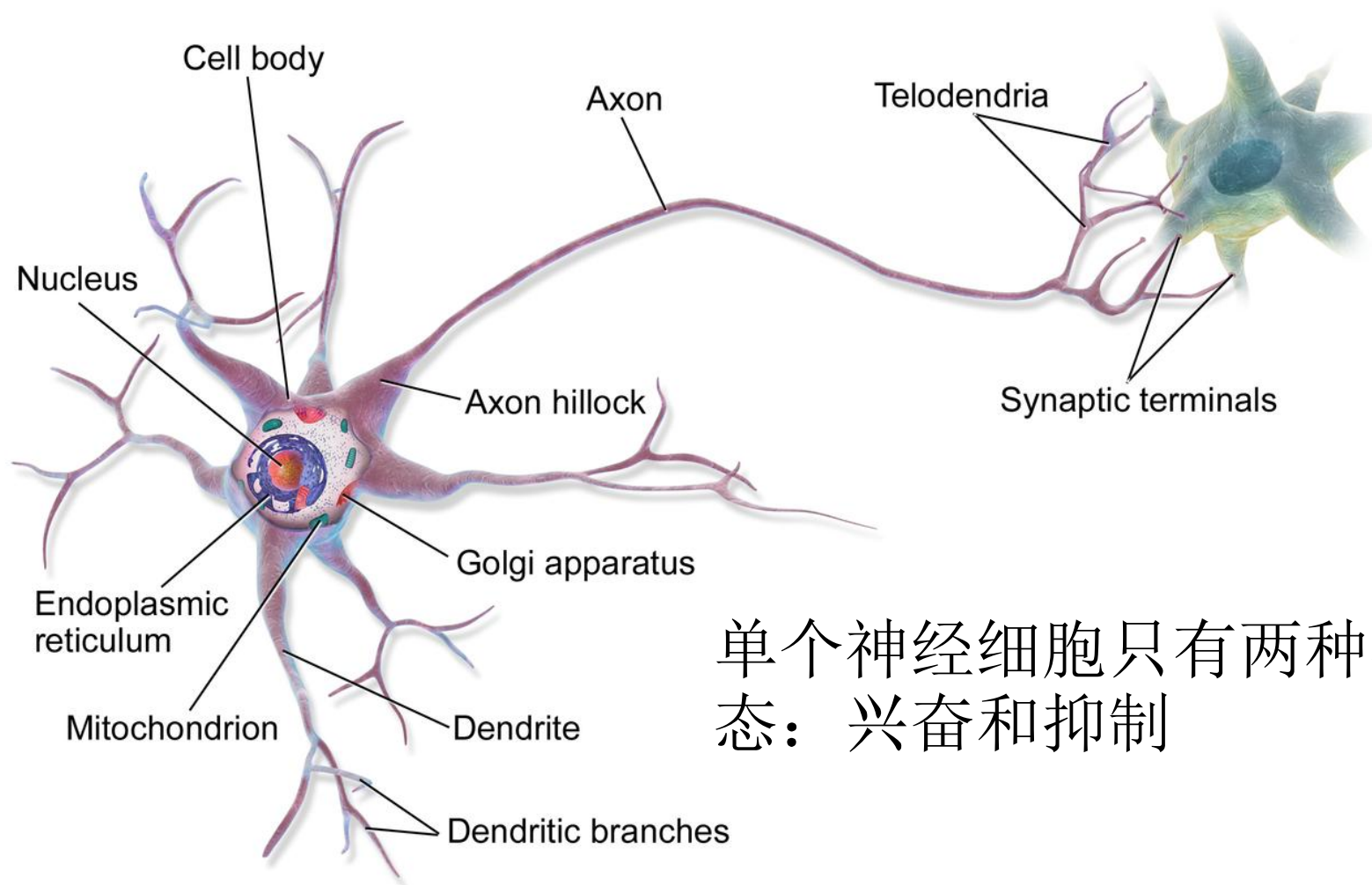
Image credit: Alec Radford

为什么现在才显式威力？

- 缺点
 - 参数过多
 - 影响训练
 - 非凸优化问题
 - 存在局部最优而非全局最优解
 - 梯度消失
 - 下层参数比较难调
 - 参数解释性差

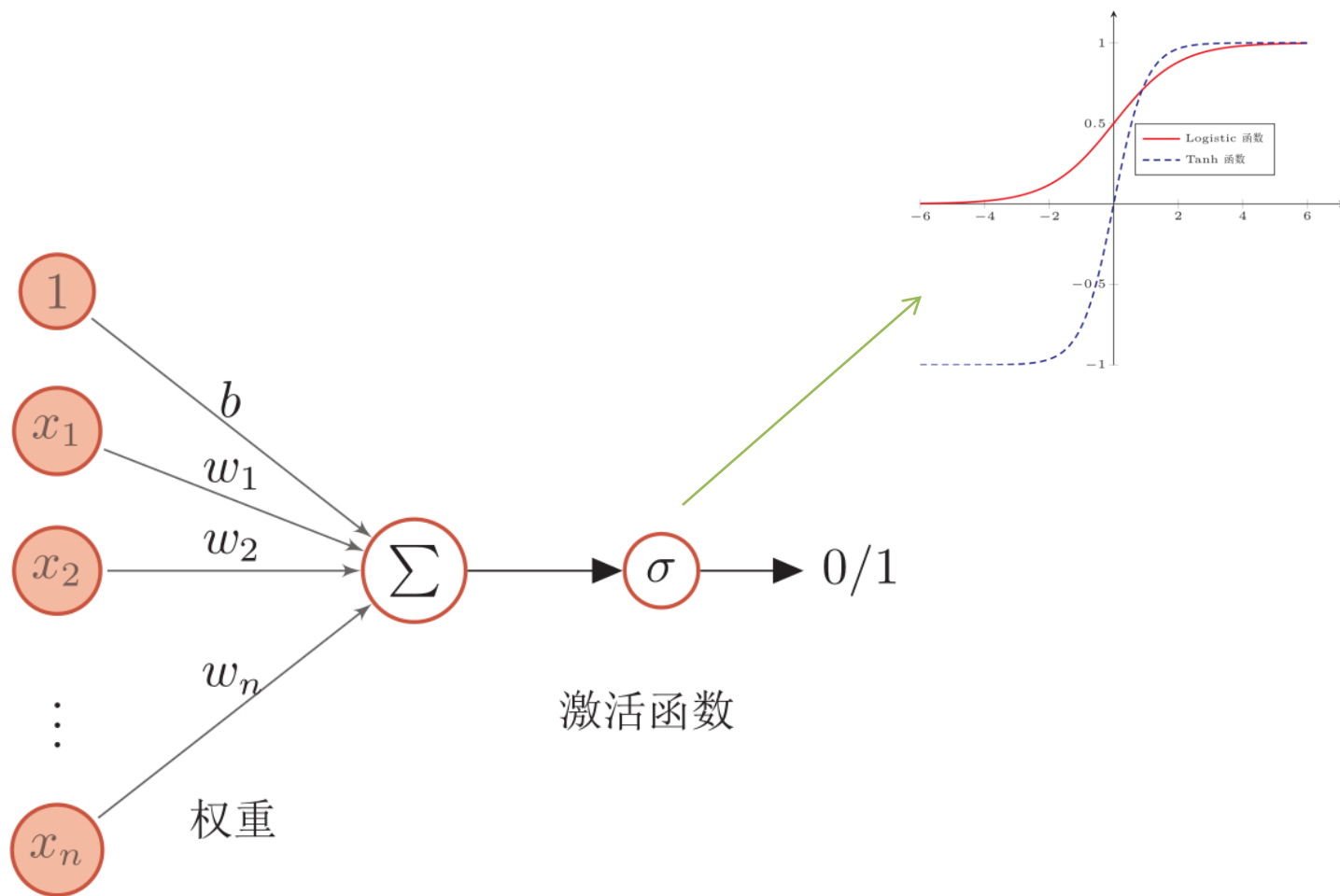


神经网络



单个神经细胞只有两种状态：兴奋和抑制

人工神经元

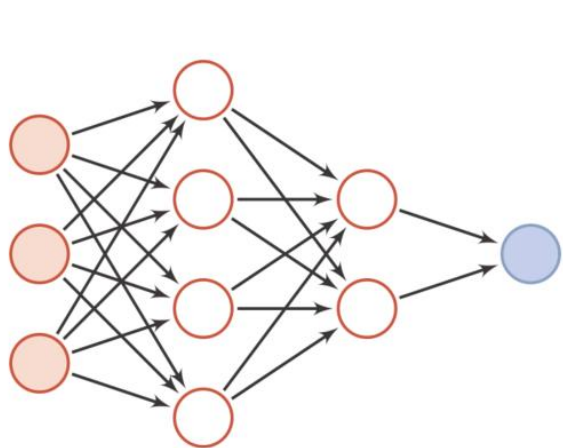


人工神经网络

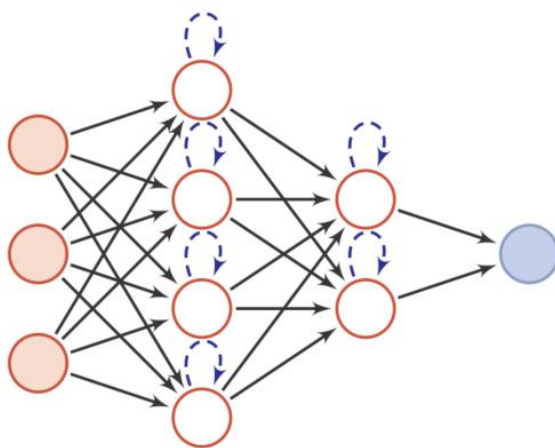
- 人工神经网络主要由大量的神经元以及它们之间的有向连接构成。因此考虑三方面
- 神经元的激活规则
 - 主要是指神经元输入到输出之间的映射关系，一般为非线性函数。
- 网络的拓扑结构
 - 不同神经元之间的连接关系。
- 学习算法
 - 通过训练数据来学习神经网络的参数。

人工神经网络

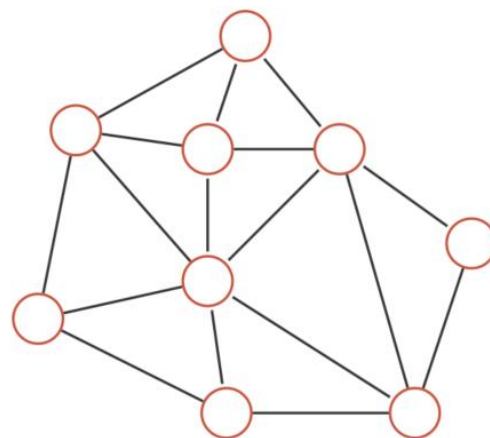
- 人工神经网络由神经元模型构成，这种由许多神经元组成的信息处理网络具有并行分布结构。



(a) 前馈网络



(b) 反馈网络



(c) 图网络

卷积神经网络

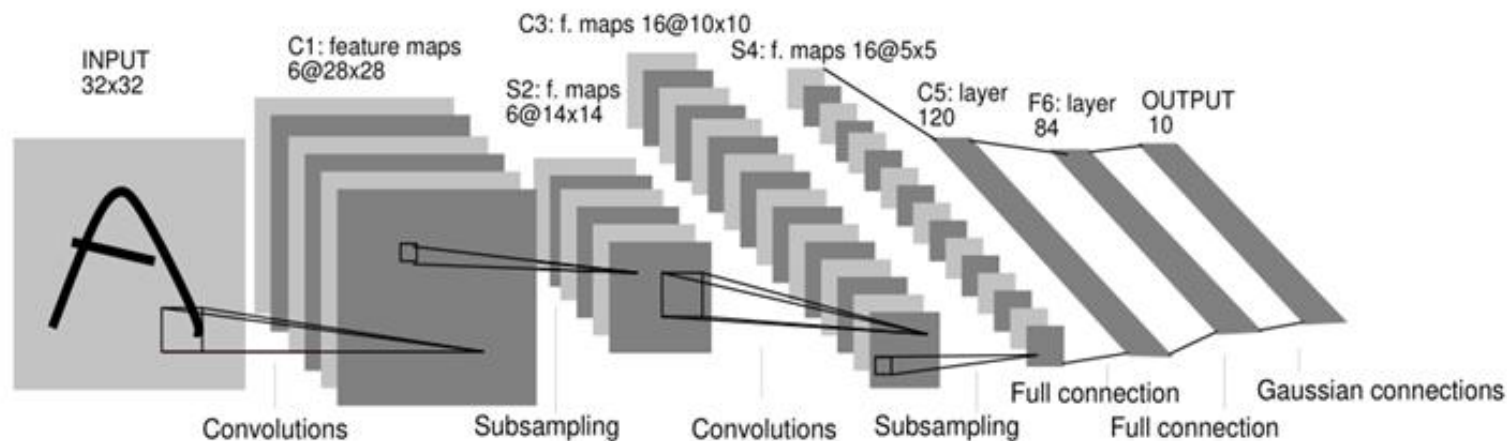
LeNet

CNN的演变——LeNet

1. LeNet

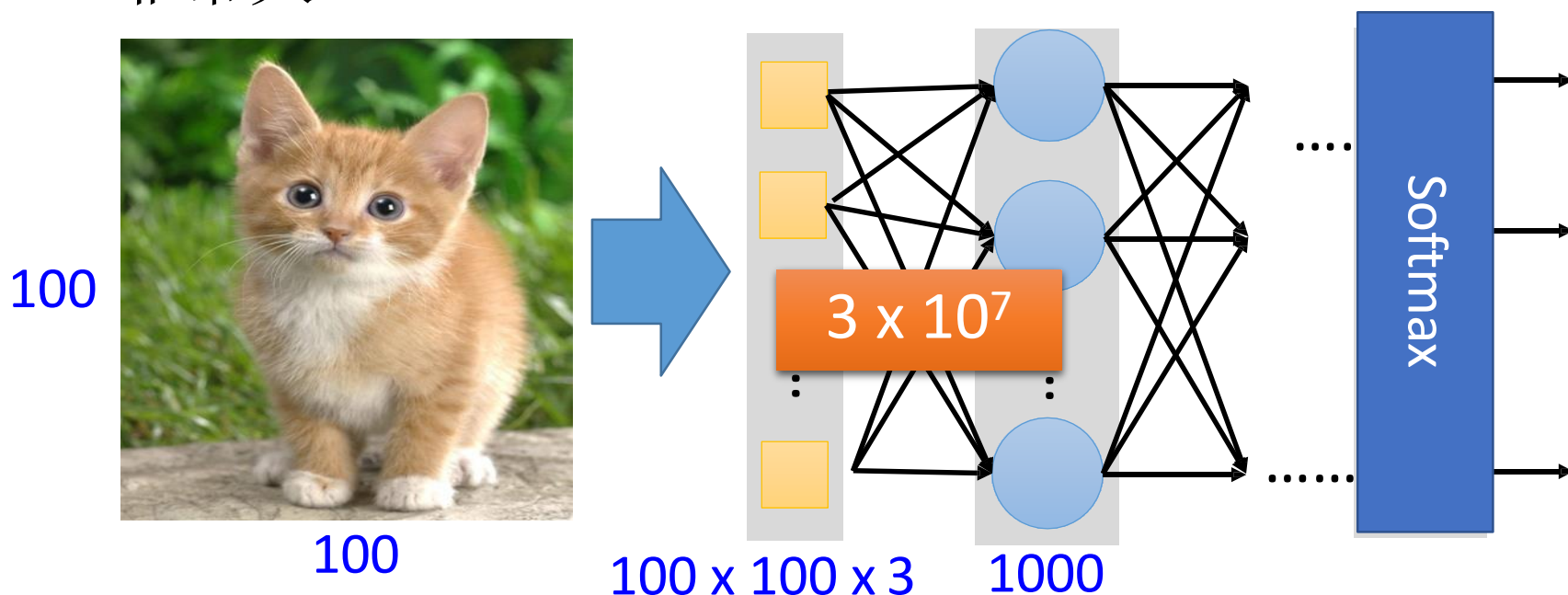
LeNet 的论文详见 http://vision.stanford.edu/cs598_spring07/papers/Lecun98.pdf。LeNet 包含的组件如下。

- 输入层： 32×32 。
- 卷积层：3 个。
- 降采样层：2 个。
- 全连接层：1 个。
- 输出层（高斯连接）：10 个类别（数字 0~9 的概率）。



为什么CNN 用于图像识别?

- 当处理图像时，全连接网络的第一层将会非常大.....



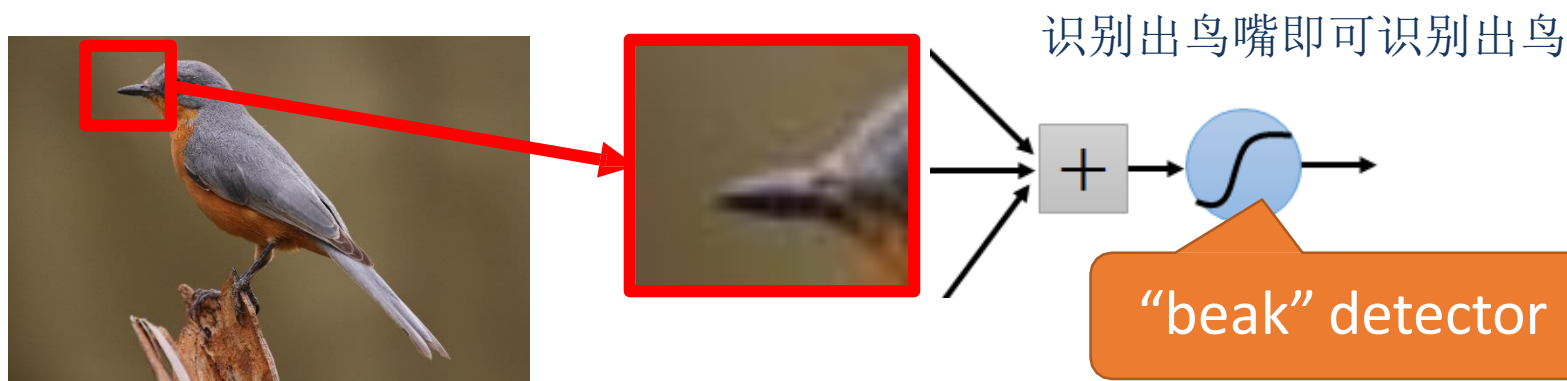
可以通过考虑图像识别的性质来简化全连接网络吗?

为什么CNN 用于图像识别?

- 一些模式比整张图片小得多

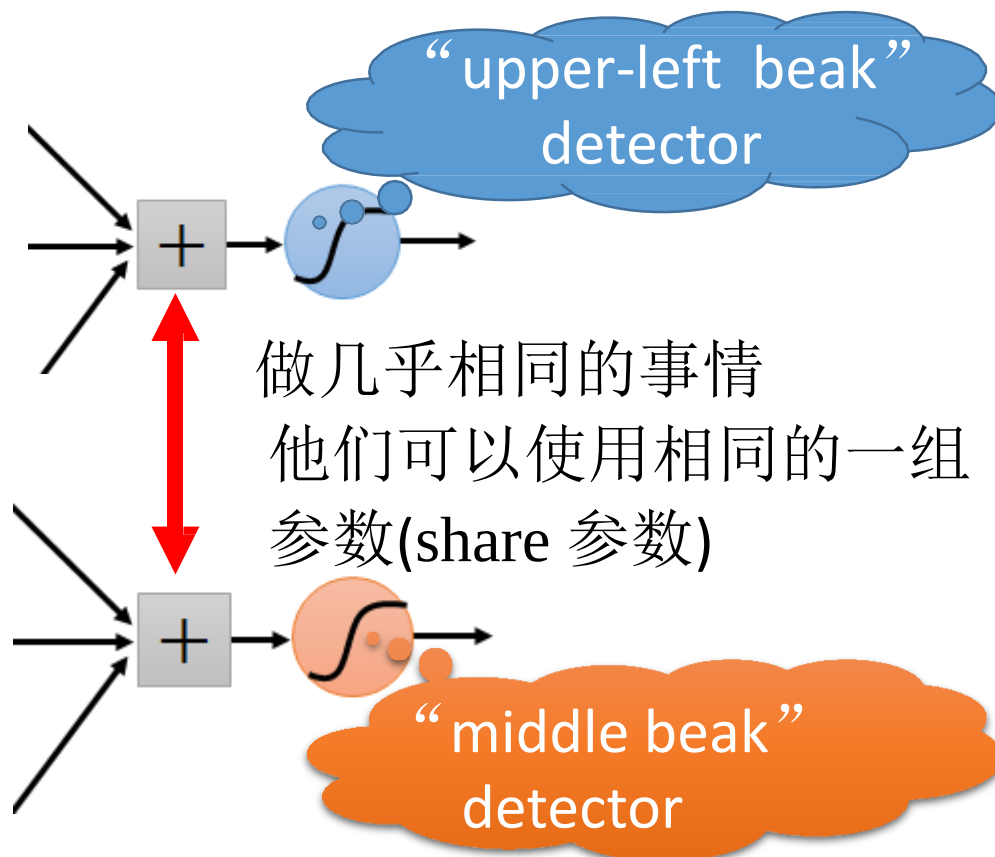
一个神经元不需要看到整个图像去发现模式

通过较少的参数连接到小区域



为什么CNN 用于图像识别?

- 同样的模式可能出现在图像的不同区域



为什么CNN 用于图像识别?

- 对图像进行二次采样不会改变图像中的物体



二次采样



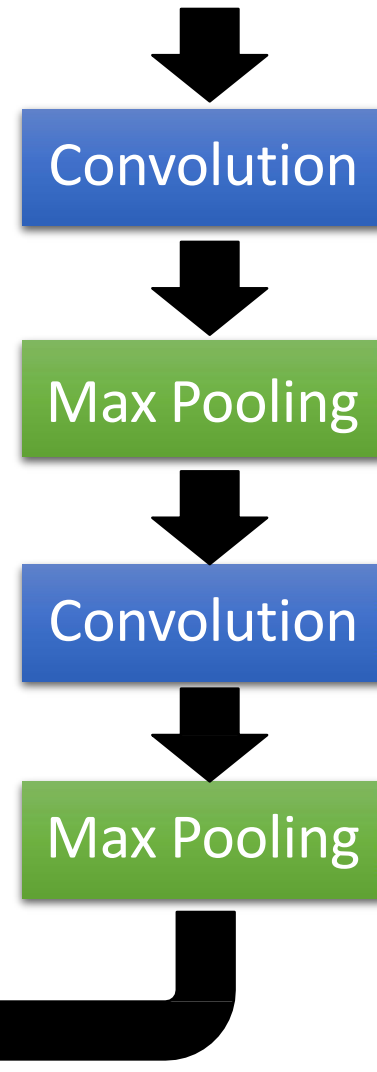
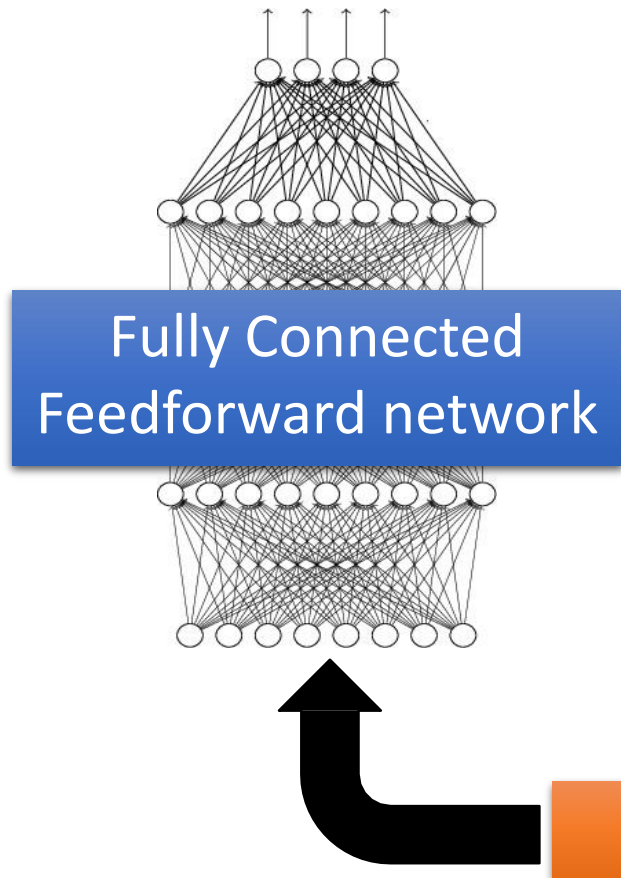
我们可以对图像进行二次采样以使图像变小



更少的参数来处理图像

The whole CNN

cat dog



可以重复
多次

The whole CNN

属性 1

- 一些模式比整张图片小得多

属性 2

- The 同样的模式可能出现在图像的不同区域

属性 3

- 对图像的二次采样不会改变图像中的物体



Convolution

Max Pooling

Convolution

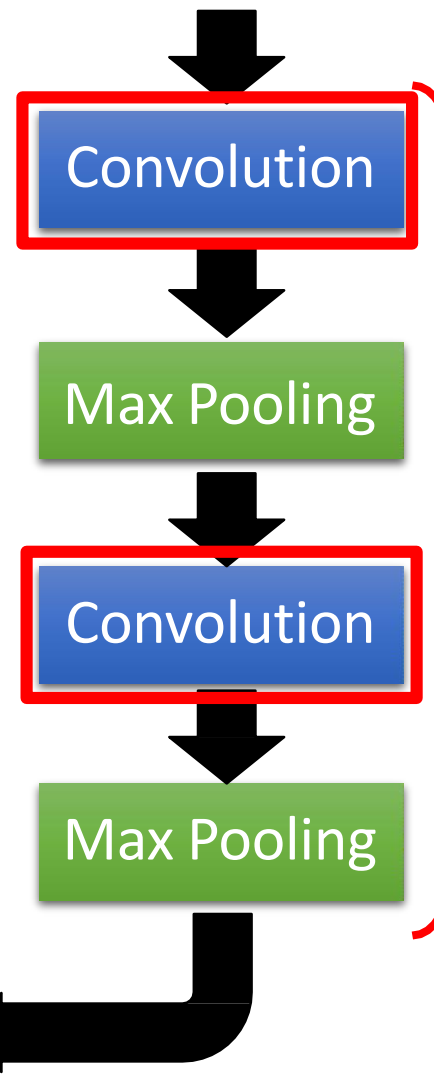
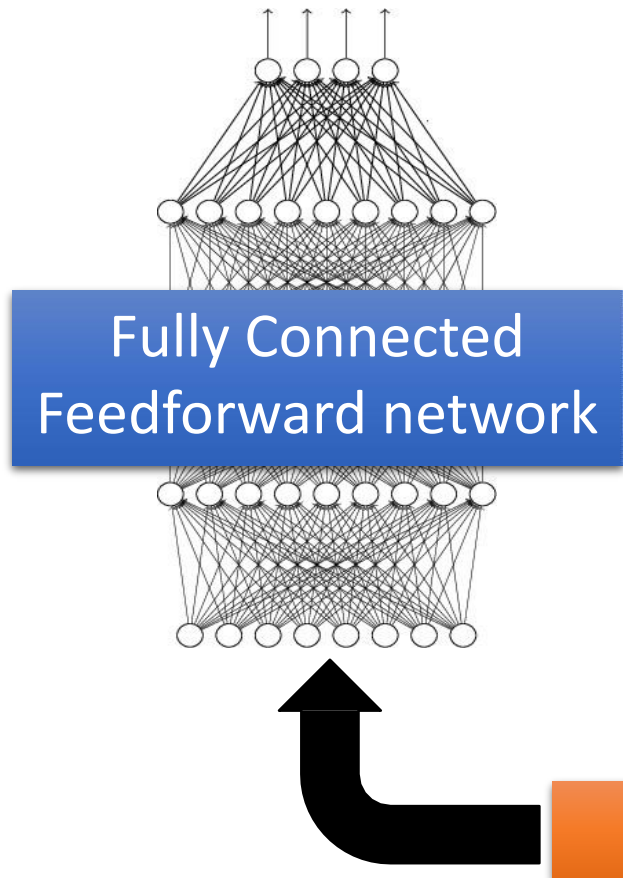
Max Pooling

可以重复多次

Flatten

The whole CNN

cat dog



可以重复
多次

CNN – 卷积

这些都是要学习的参数

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1
矩阵

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2
矩阵

⋮

Property 1

每个 filter 探测一个小的模式(3 x 3).

来自李宏毅的深度学习教程

CNN – 卷积

步长=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3 -1

CNN – Convolution

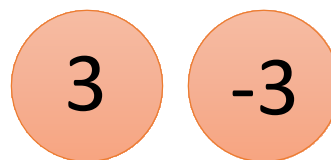
If 步长=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



We set stride=1 below

CNN – Convolution

步长=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-3	-1	-3	-1
-3	-1	0	-3
-3	-3	0	1
-3	-2	-2	-1

Property 2

CNN – Convolution

步长=1

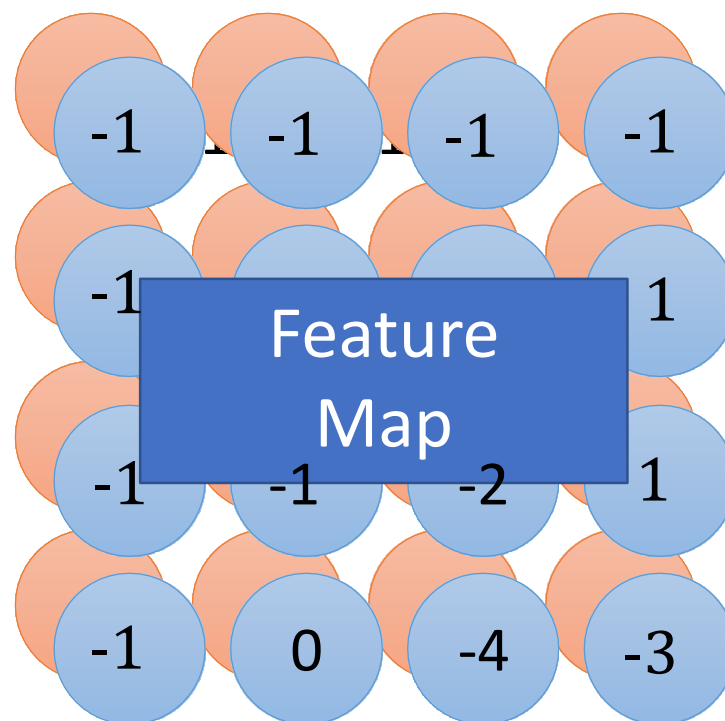
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

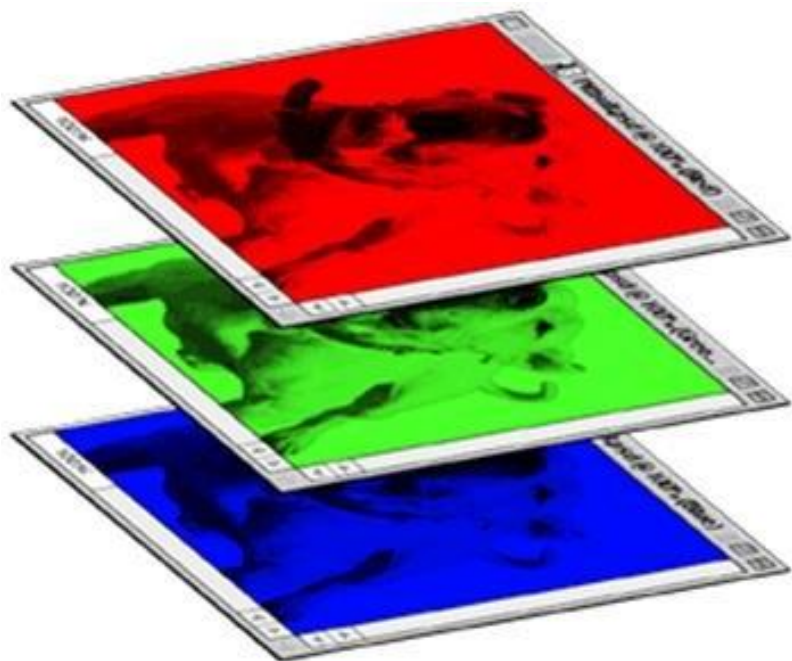
对于每一个 filter 做同样的事



4 x 4 image

CNN – 彩色图像

彩色图像



1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

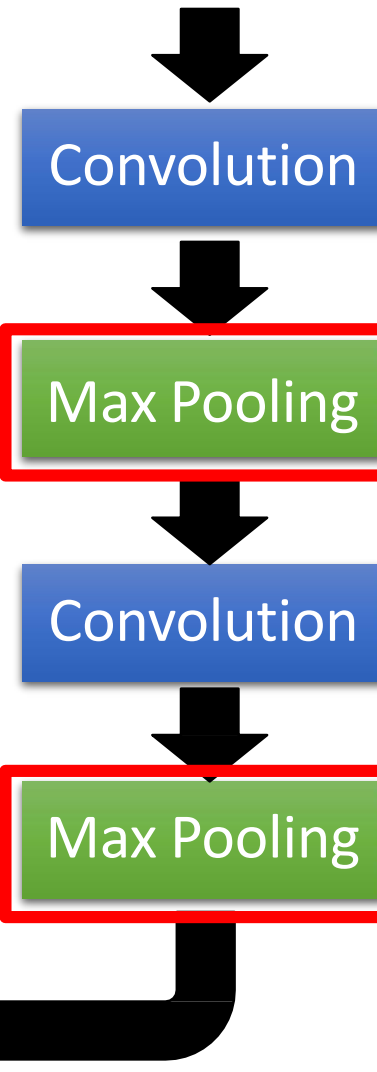
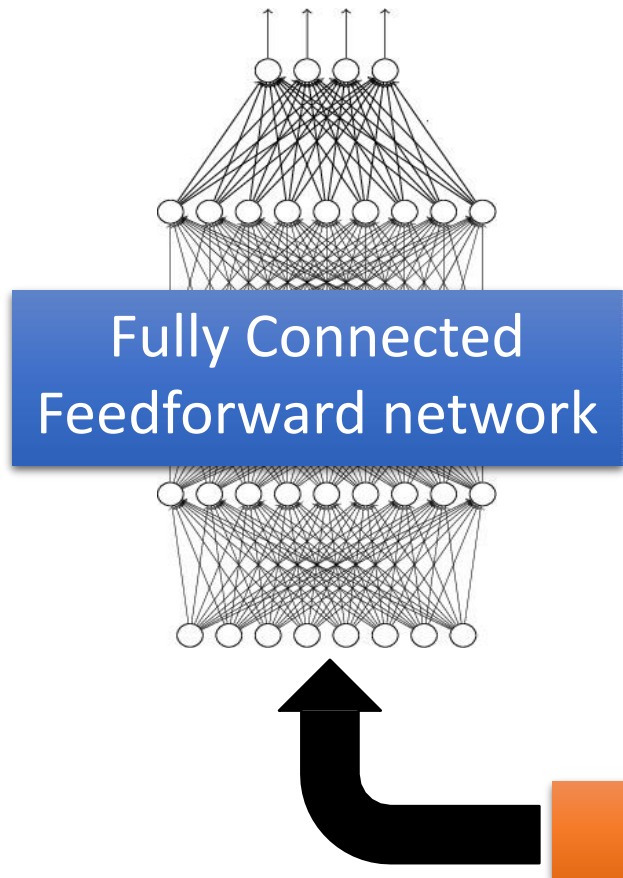
Filter 2



1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

The whole CNN

cat dog



Can repeat many times

CNN – Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

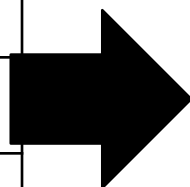
3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

CNN – Max Pooling

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

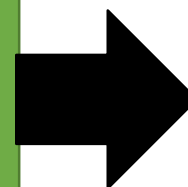
6 x 6 image



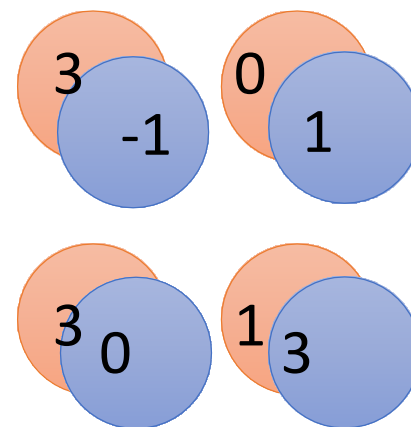
Conv



Max
Pooling

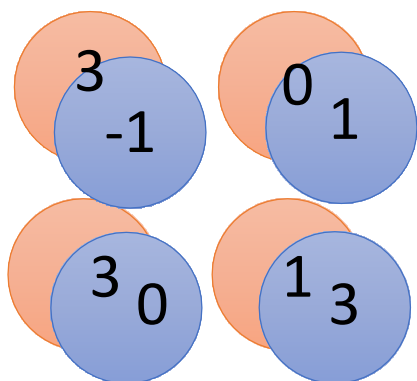


New image
but smaller



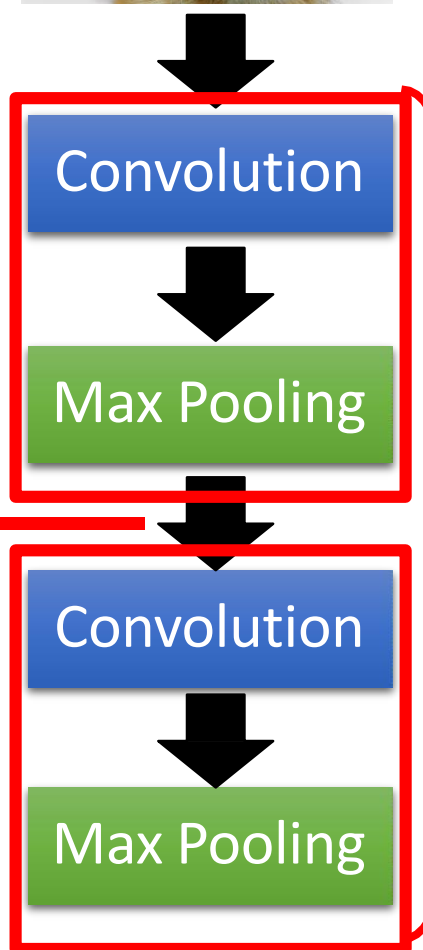
2 x 2 image

The whole CNN



新的图像

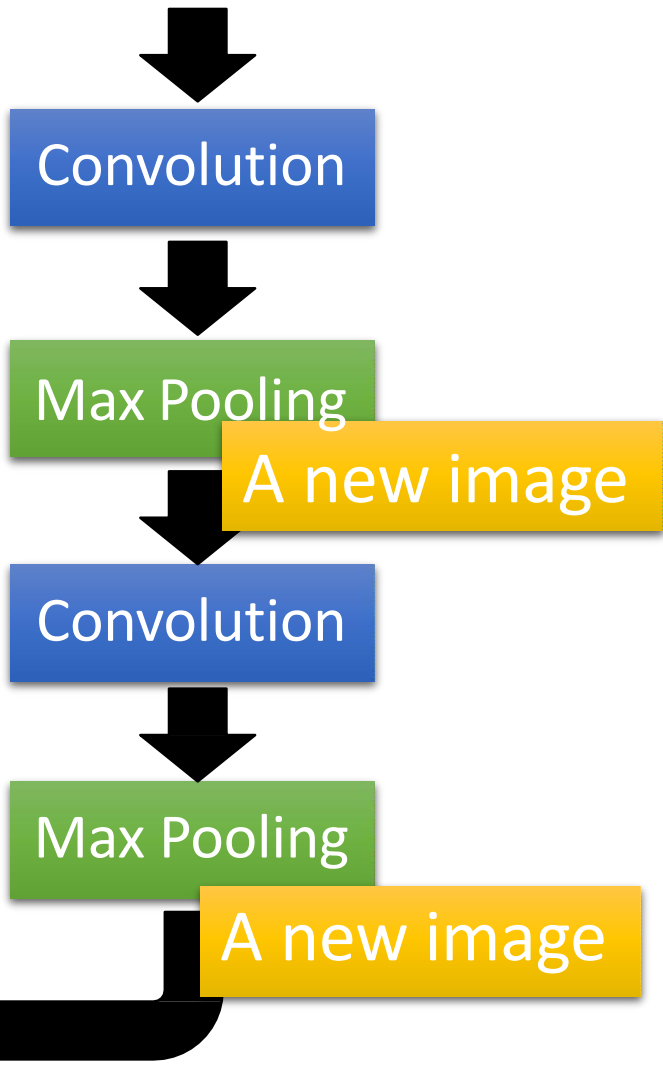
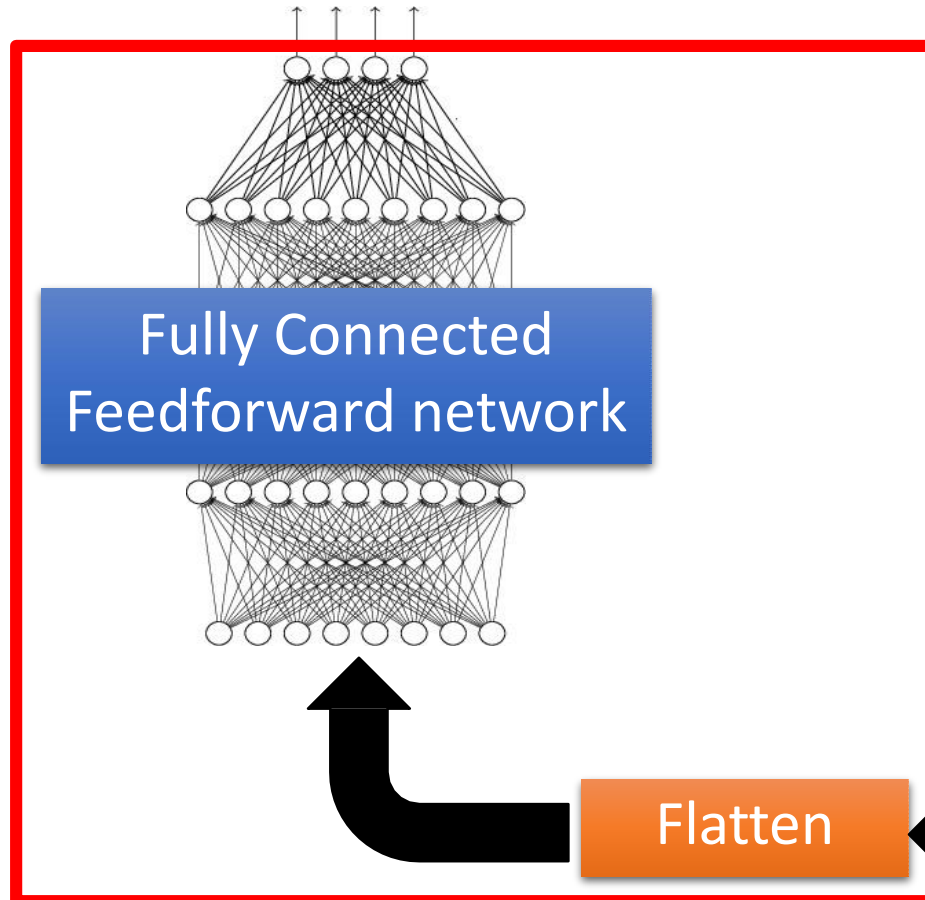
比原始图像更小



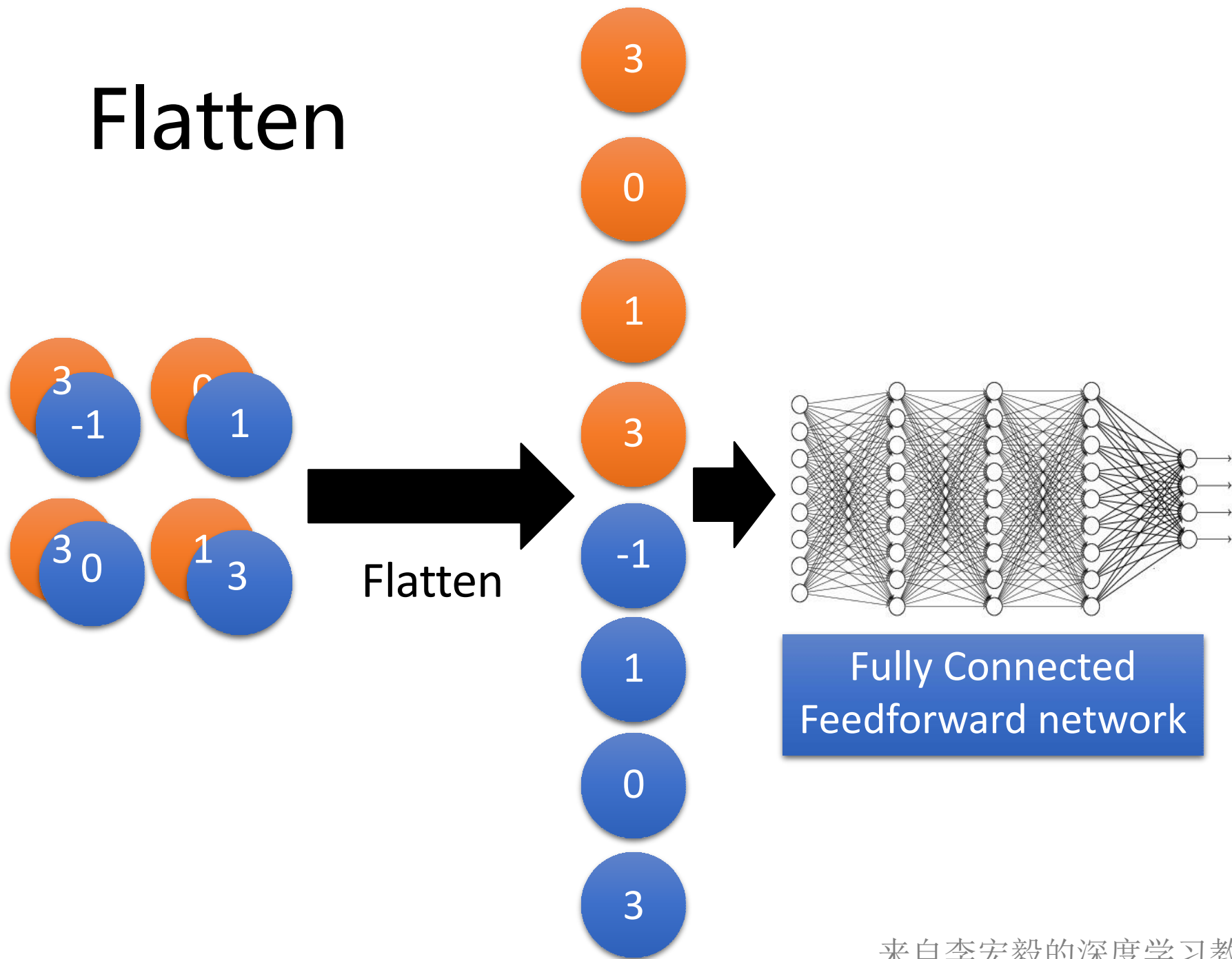
重复多次

The whole CNN

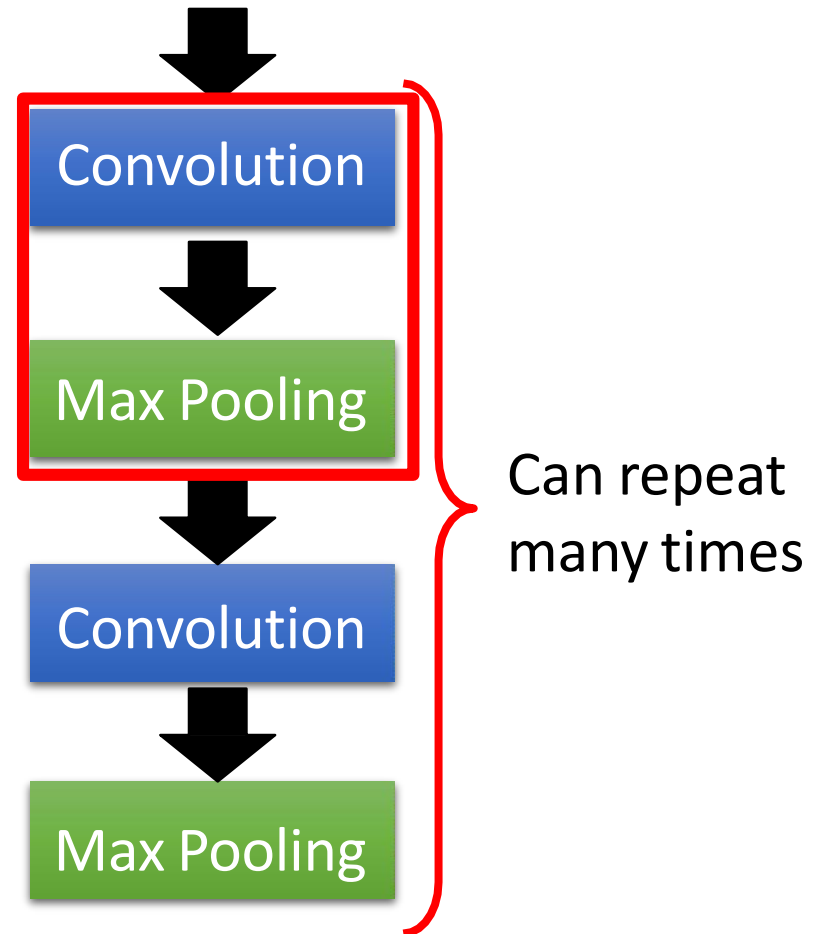
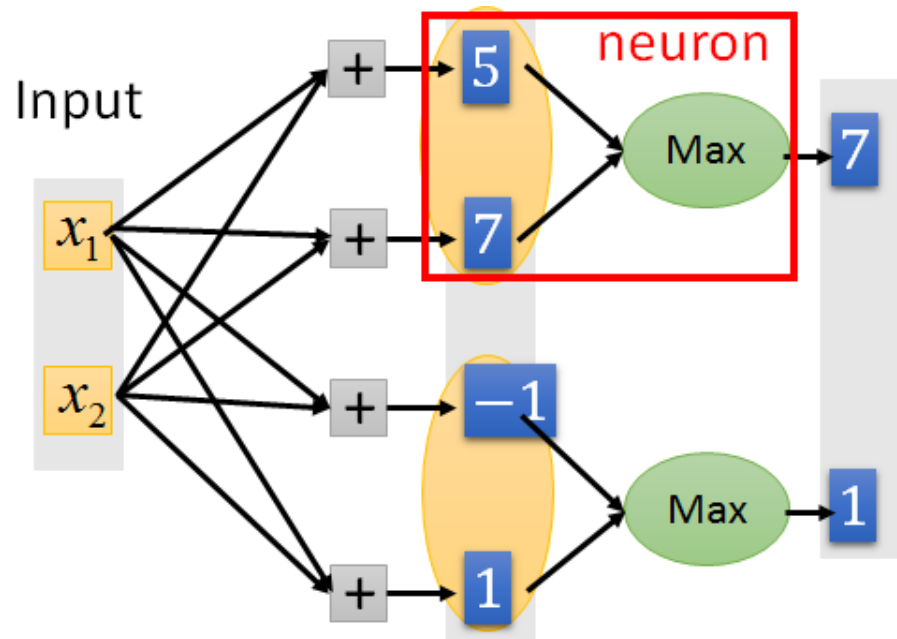
cat dog

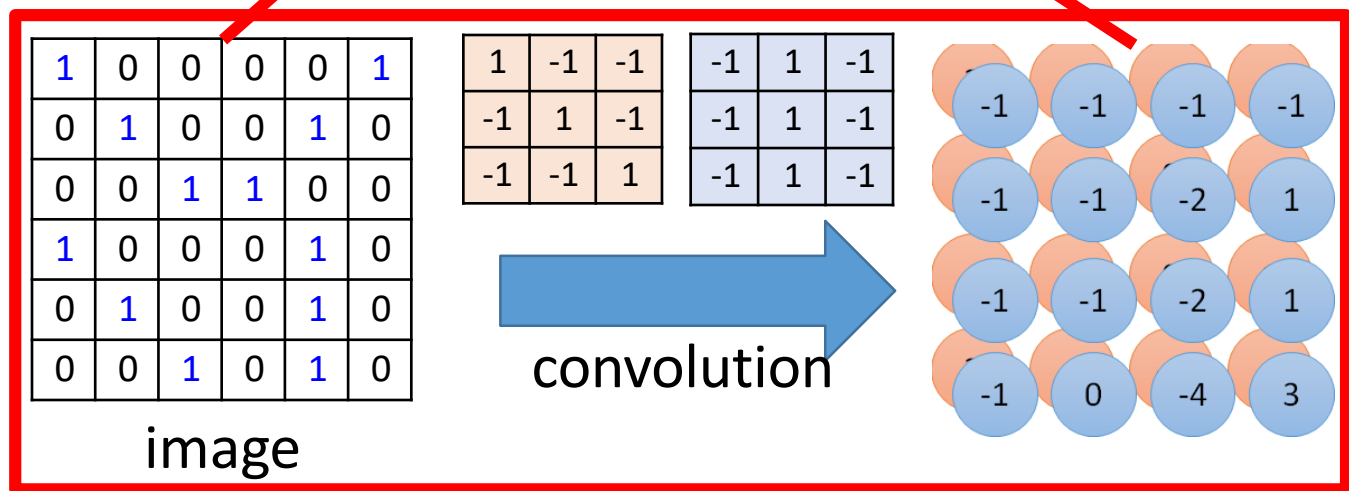
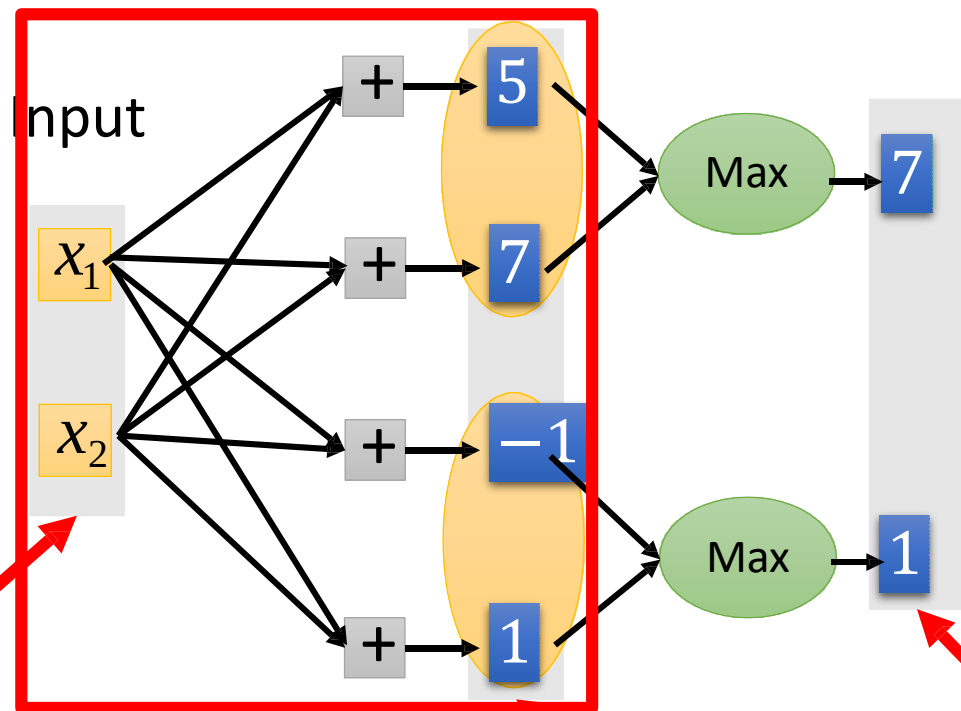


Flatten

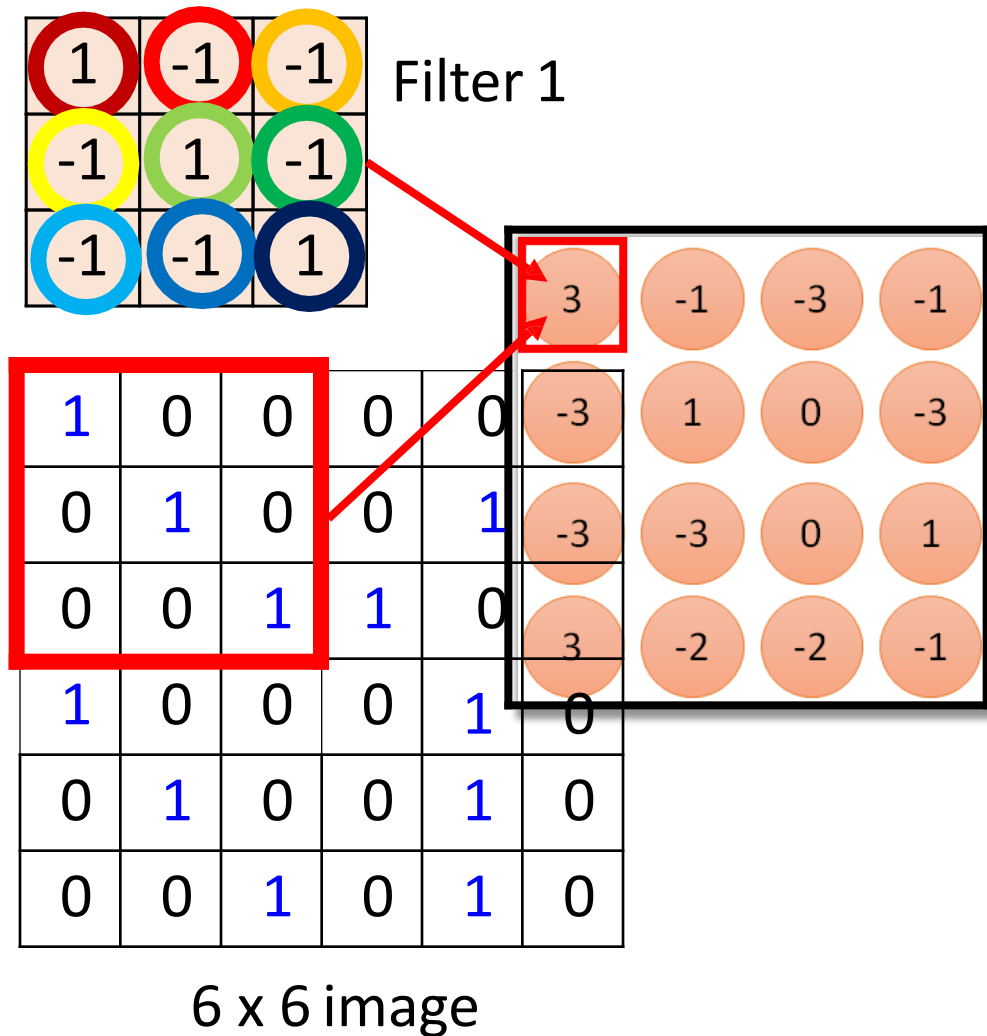


The whole CNN

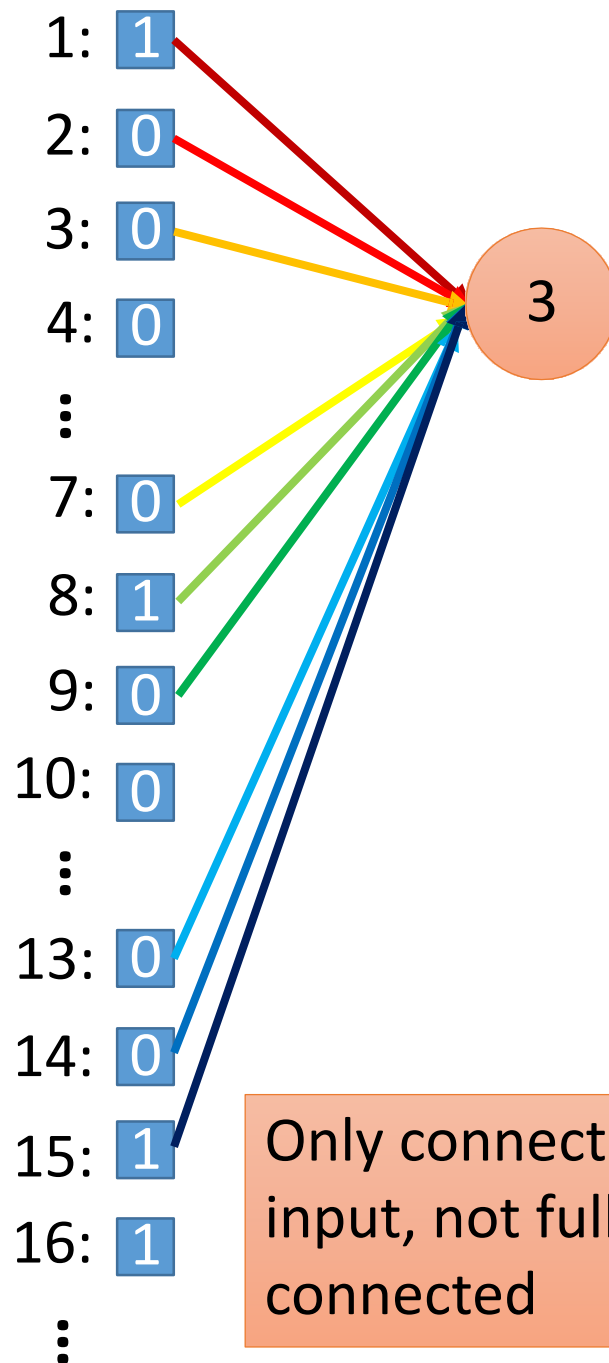


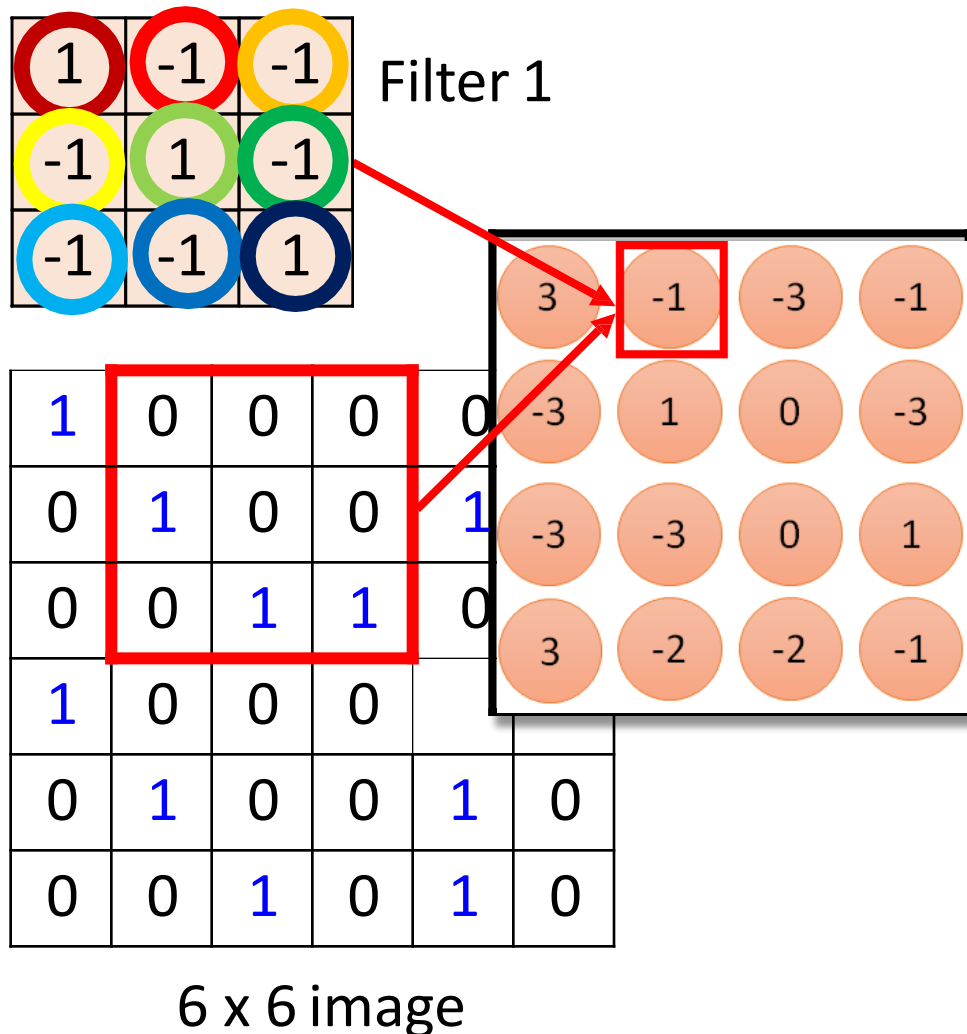


(忽略了卷积后的非线性激活函数)

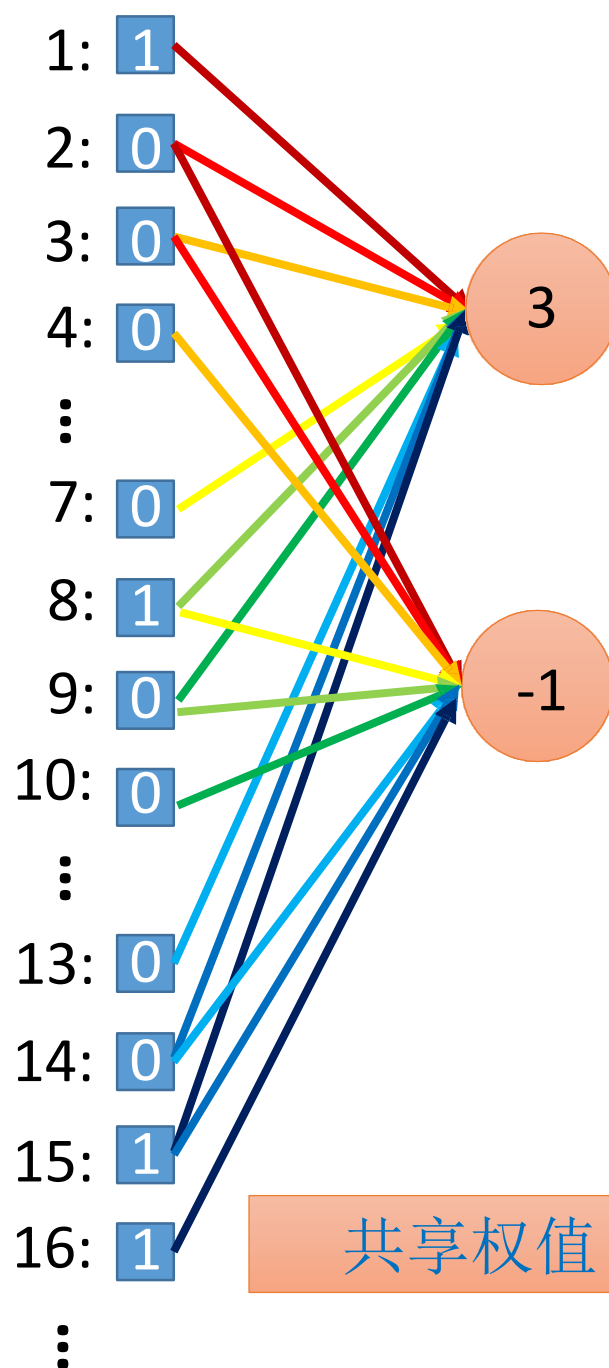


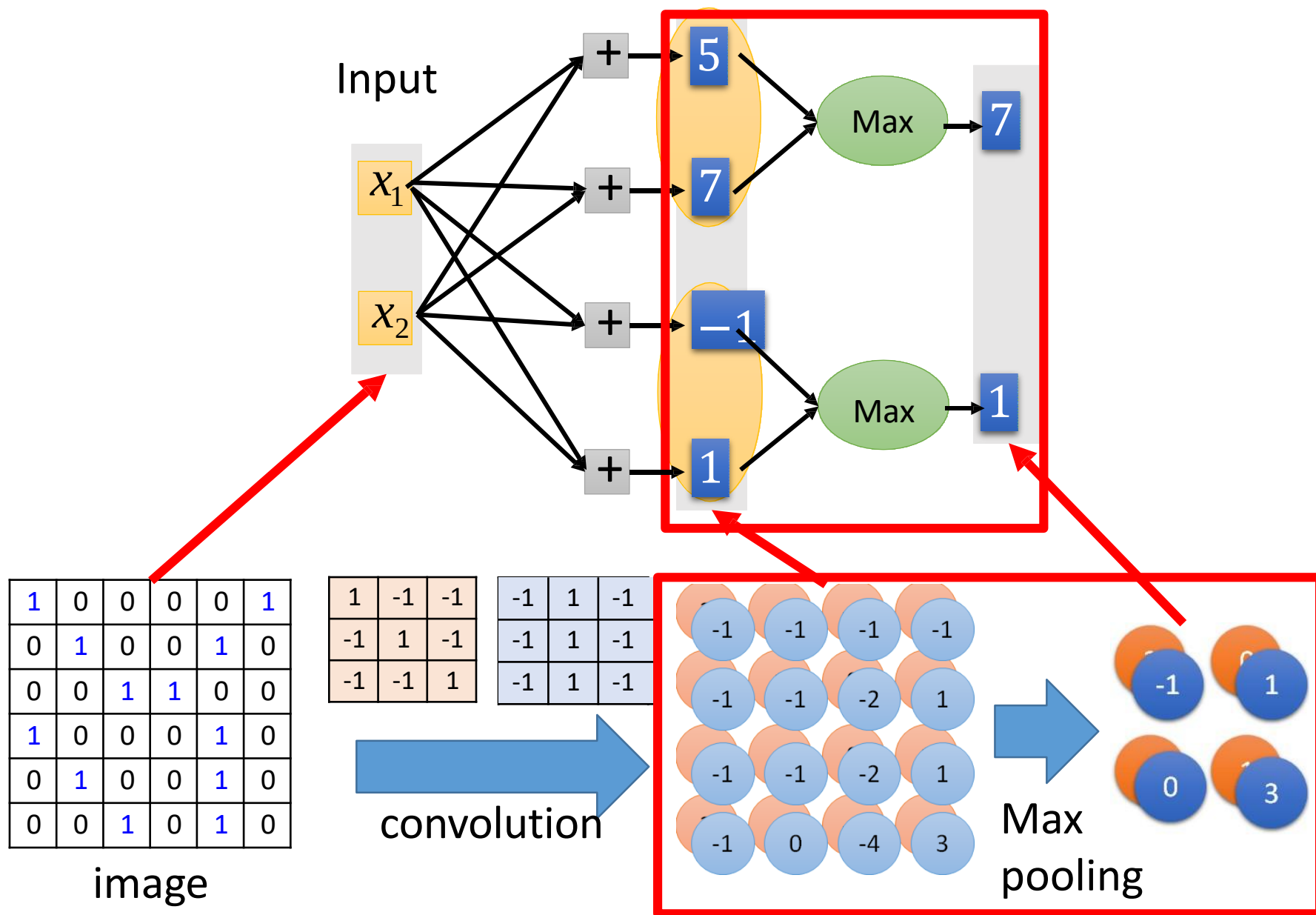
更少的参数!

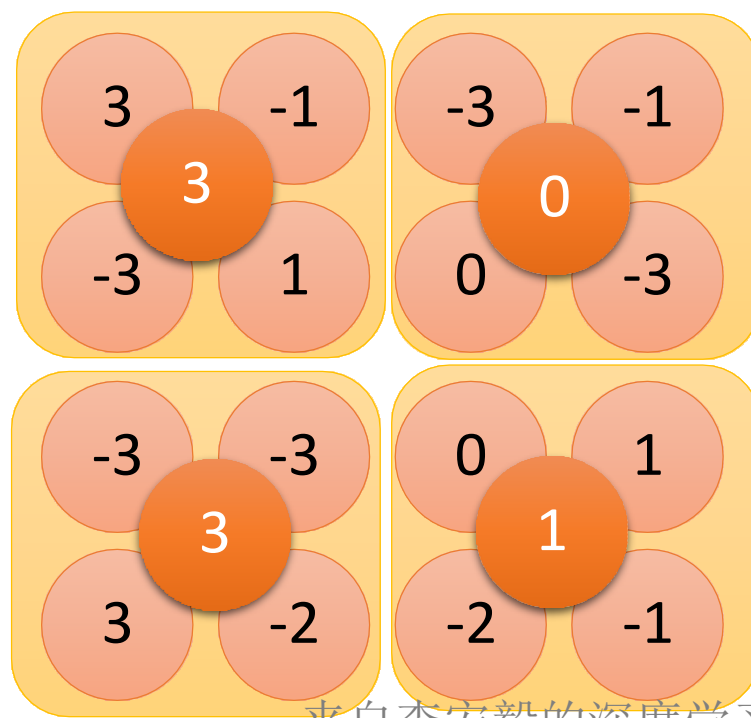
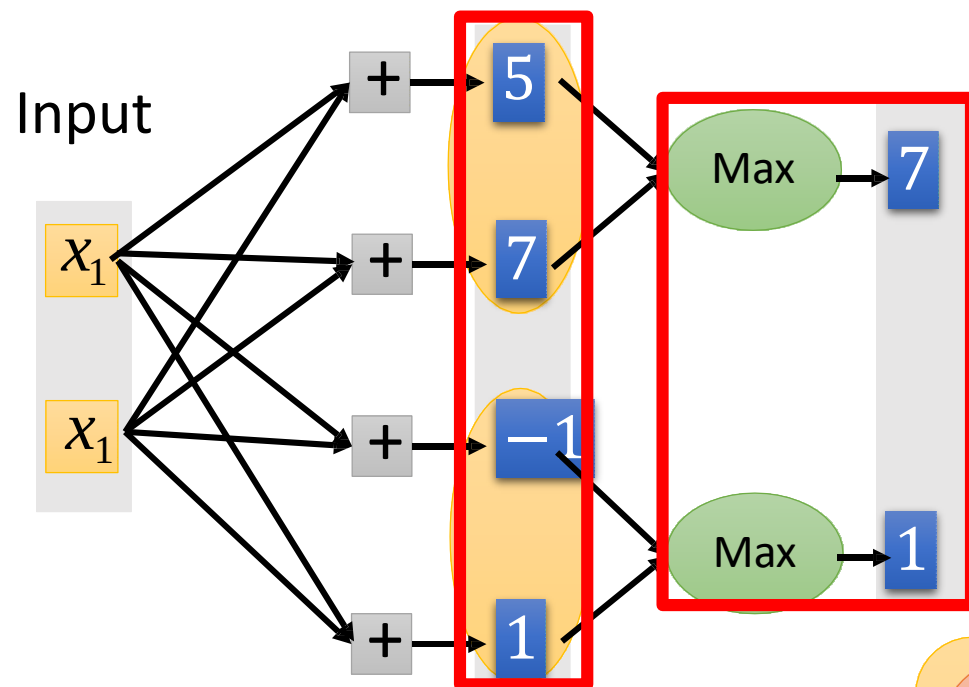




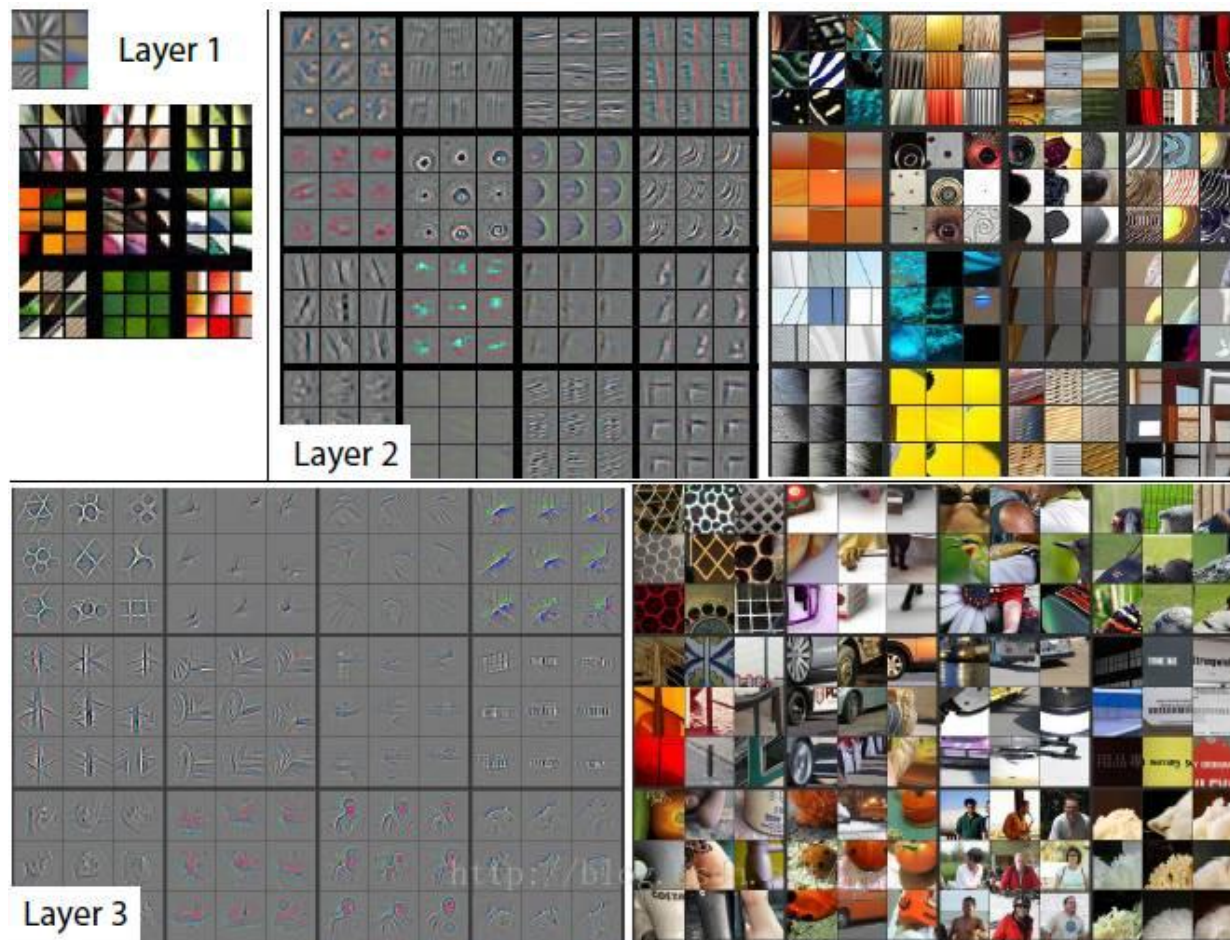
更少的参数!







- 卷积深度网络特征可视化

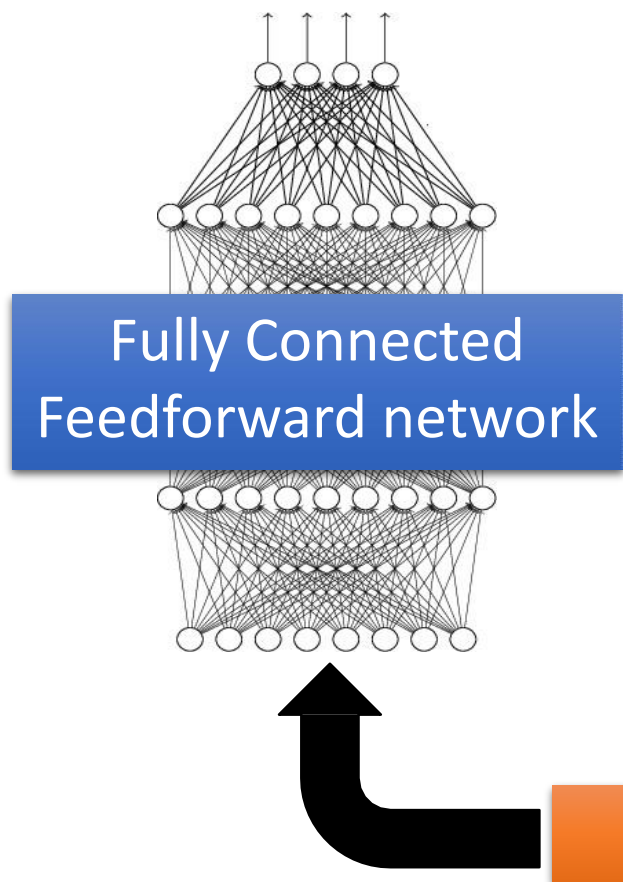


- 卷积深度网络特征可视化



The whole CNN

cat dog



Convolution

Max Pooling

Convolution

Max Pooling

Flatten

思考：
CNN特征是否具有平移、旋转、尺度不变性，对遮挡是否敏感

计算机视觉中的深度网络

图像分类

IMAGENET Large Scale Visual Recognition Challenge

Steel drum

The Image Classification Challenge:
1,000 object classes
1,431,167 images



Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle



Output:
Scale
T-shirt
Giant panda
Drumstick
Mud turtle



Russakovsky et al. arXiv, 2014

IMAGENET Large Scale Visual Recognition Challenge

Year 2010

NEC-UIUC



Dense grid descriptor:
HOG, LBP

Coding: local coordinate,
super-vector

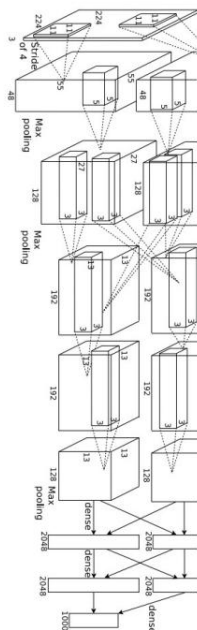
Pooling, SPM

Linear SVM

[Lin CVPR 2011]

Year 2012

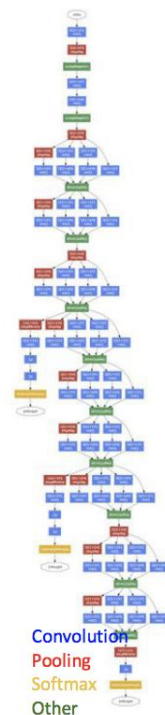
SuperVision



[Krizhevsky NIPS 2012]

Year 2014

GoogLeNet



[Szegedy arxiv 2014]

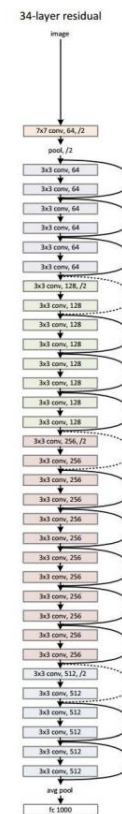
VGG



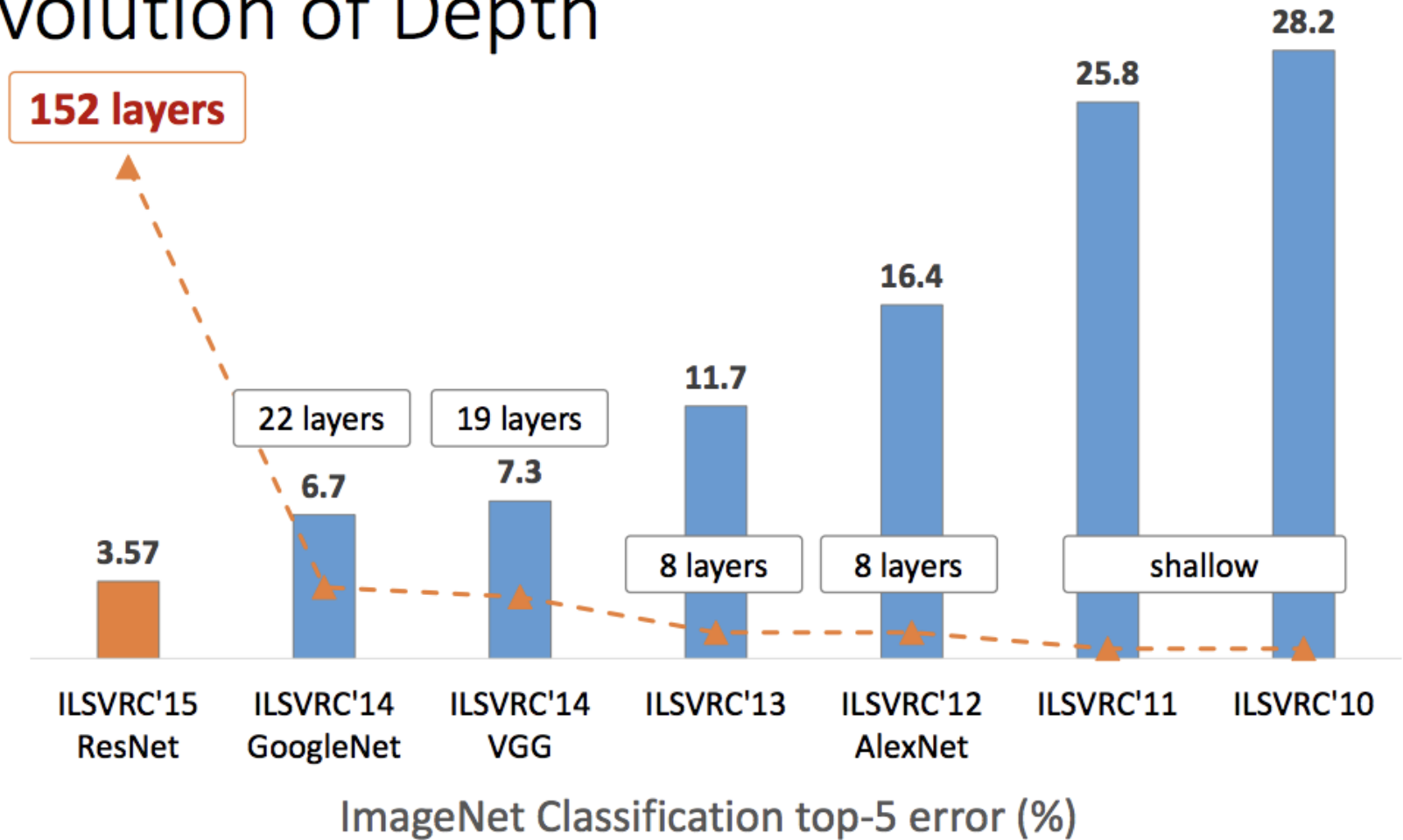
[Simonyan arxiv 2014]

Year 2015

MSRA

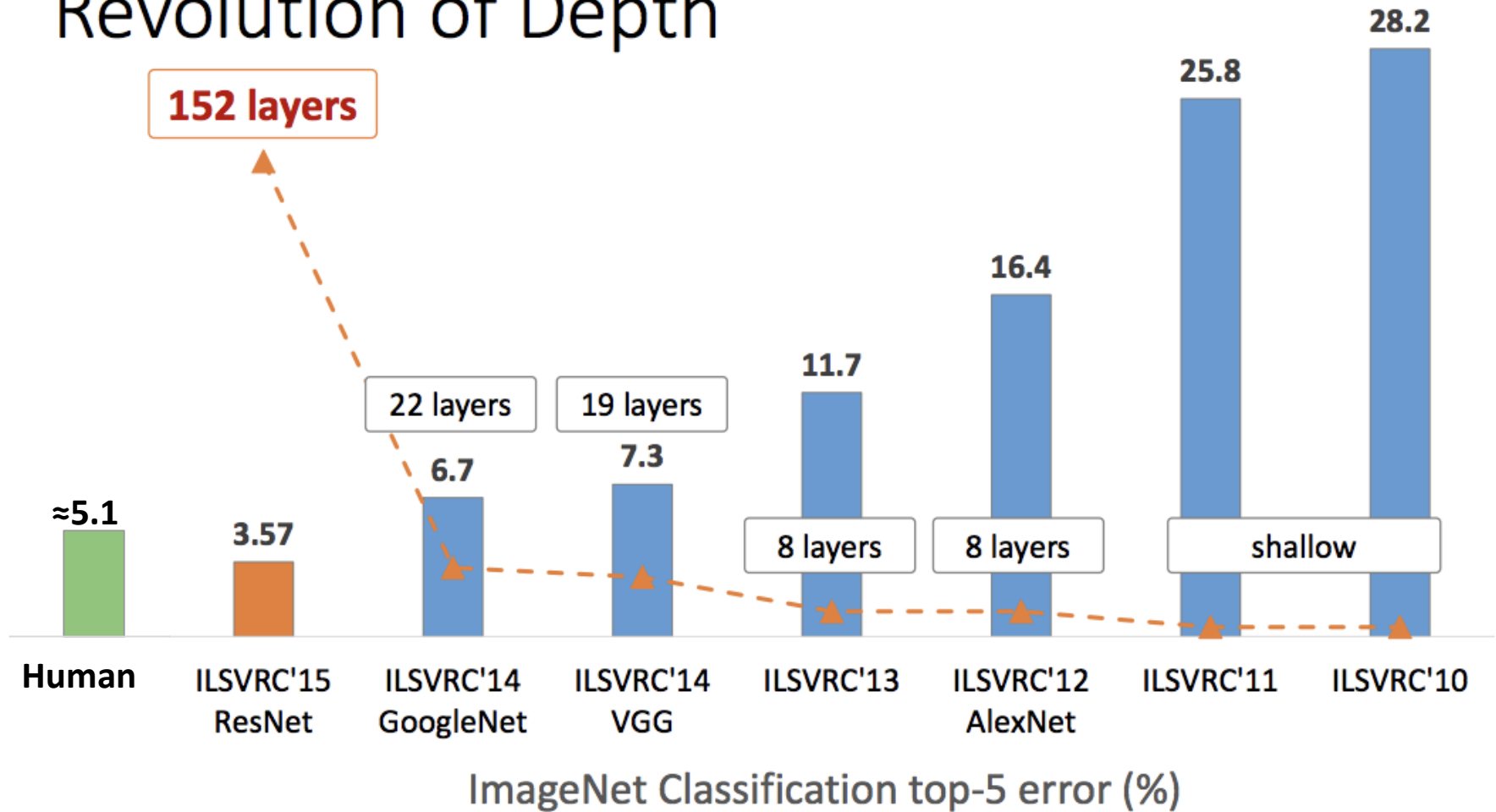


Revolution of Depth

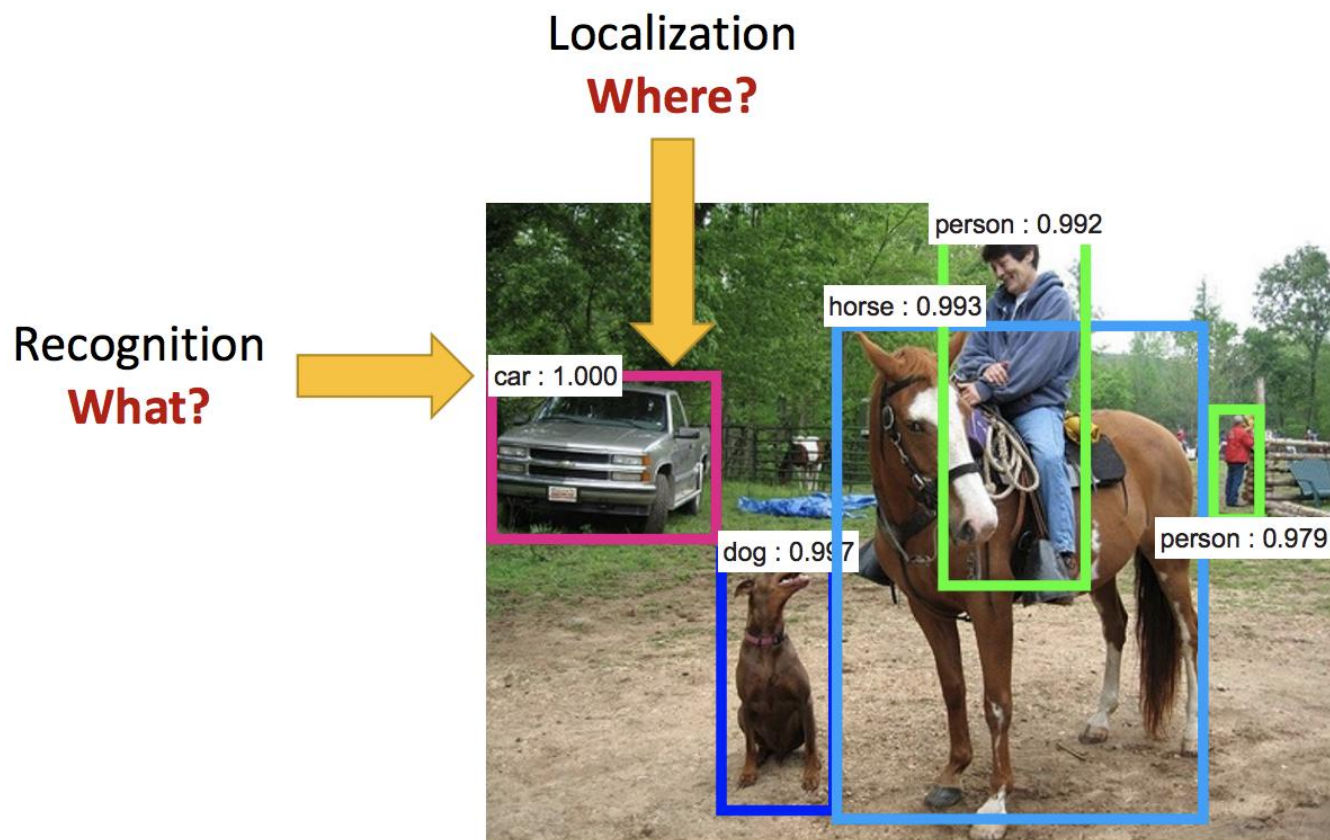


Slide credit: Kaiming He, ICCV 2015

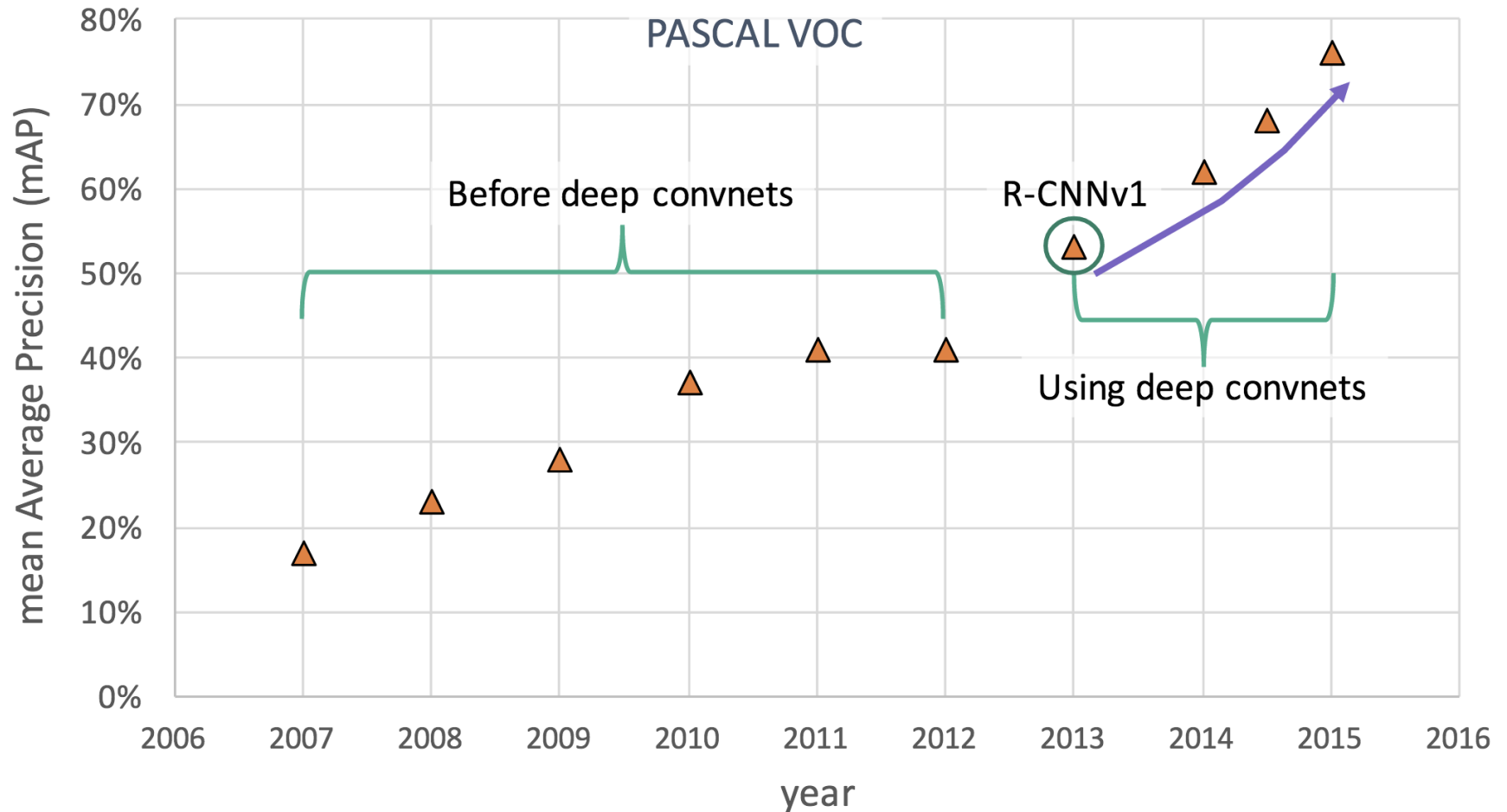
Revolution of Depth



Object Detection = What, and Where



Deep Learning for Object Detection



Object segmentation

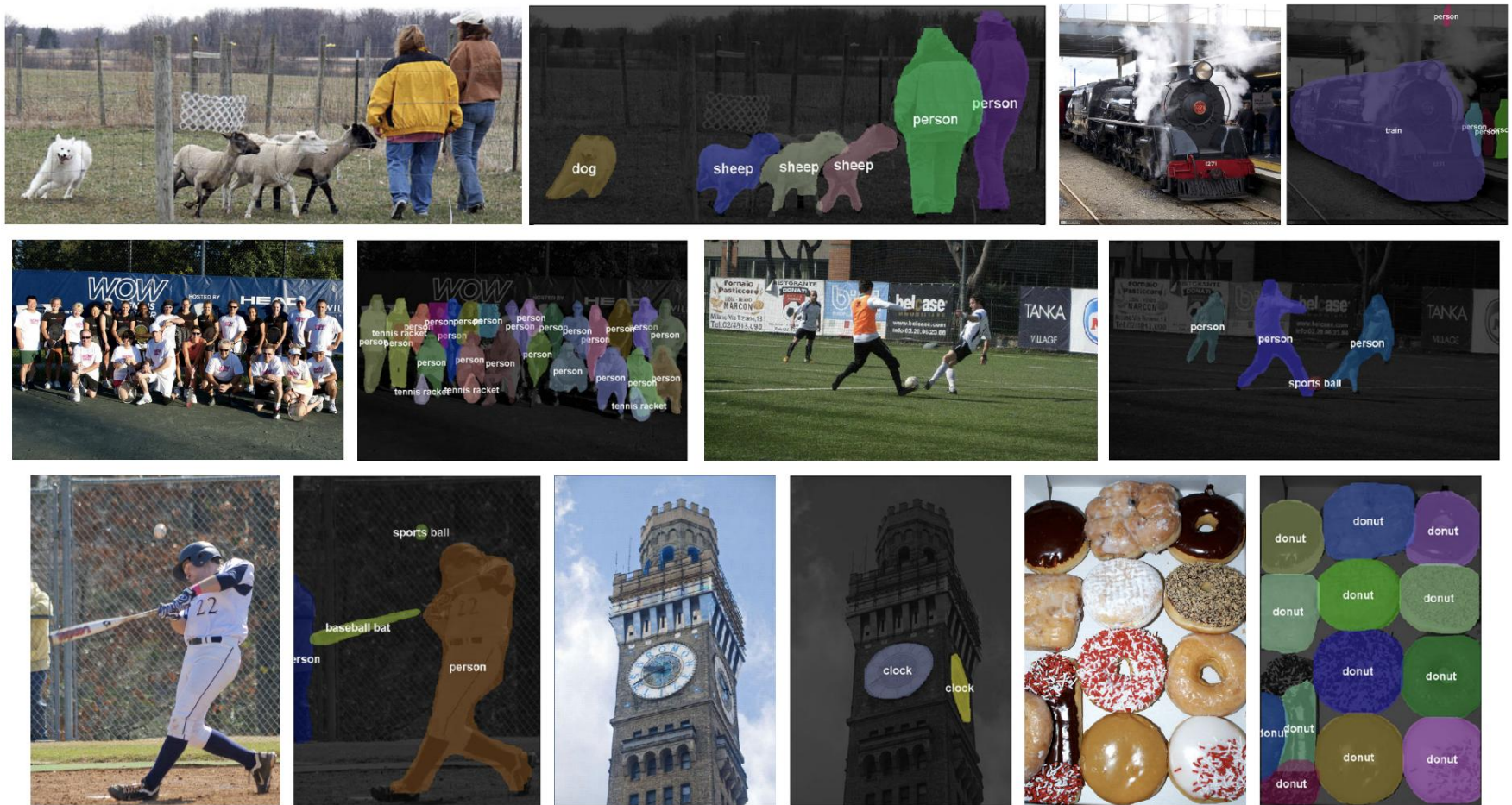


Figure credit: Dai, He, and Sun, "Instance-aware Semantic Segmentation via Multi-task Network Cascades", CVPR 2016

Pose Estimation

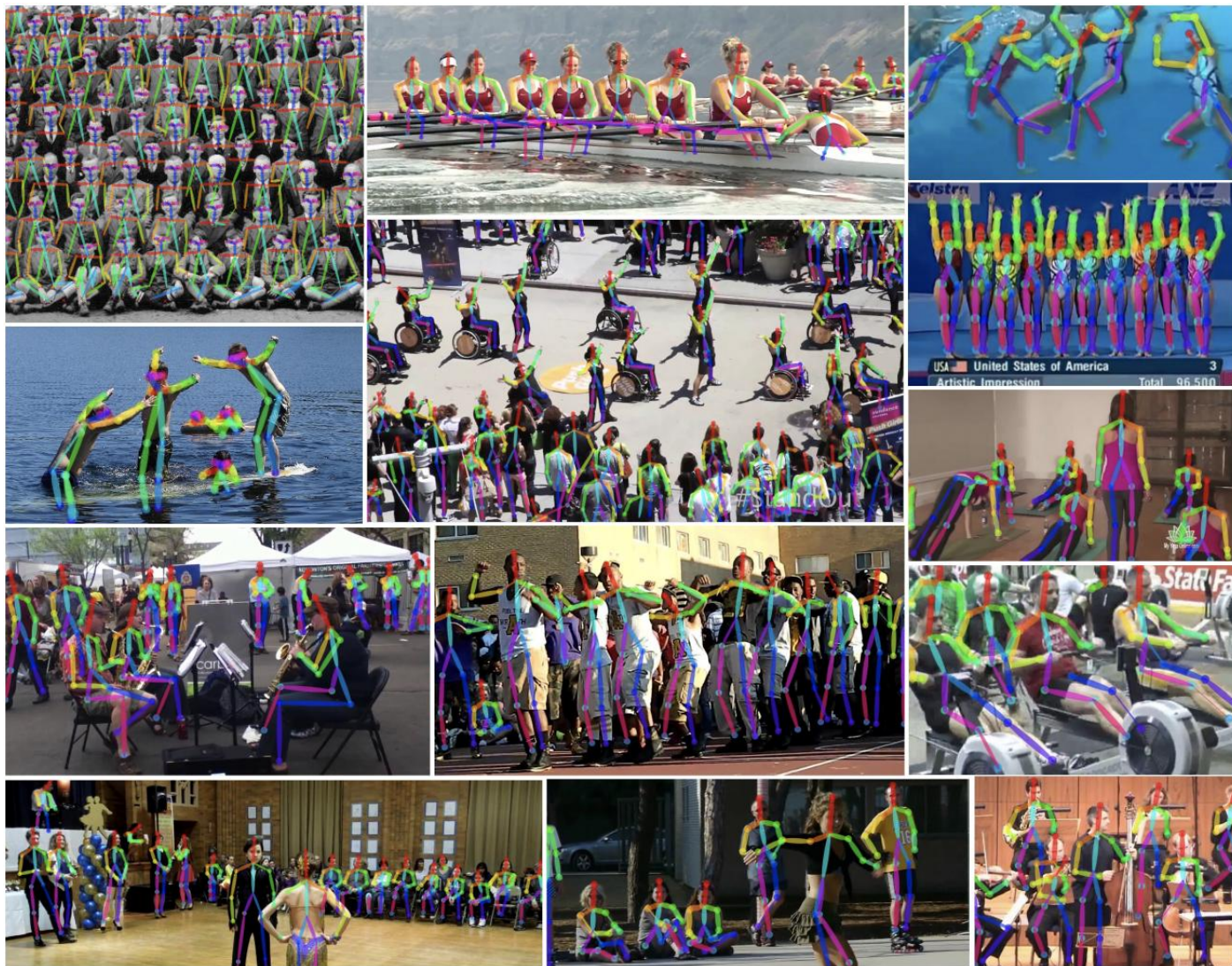


Figure credit: Cao et al, "Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields", arXiv 2016

Deep Learning for Image Captioning

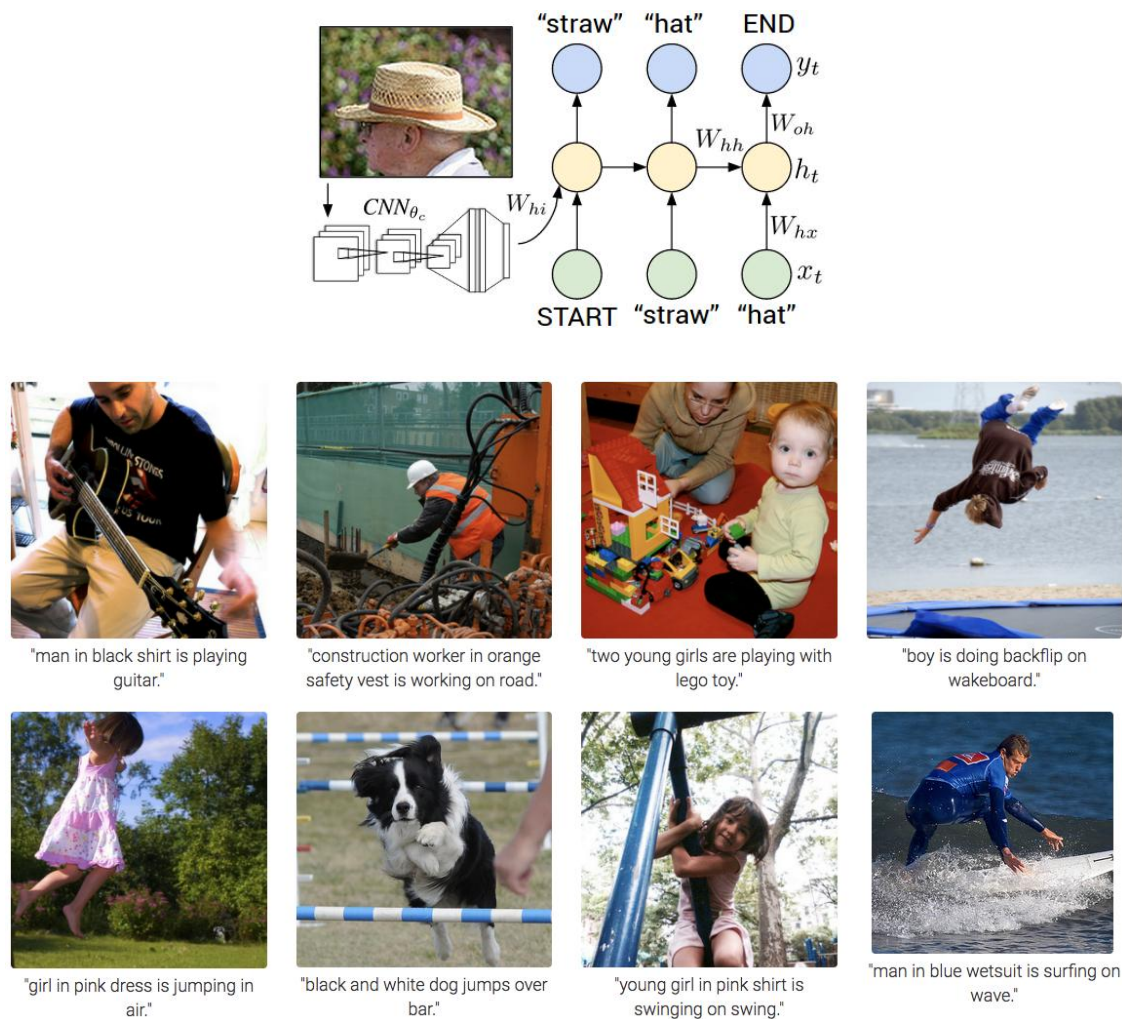
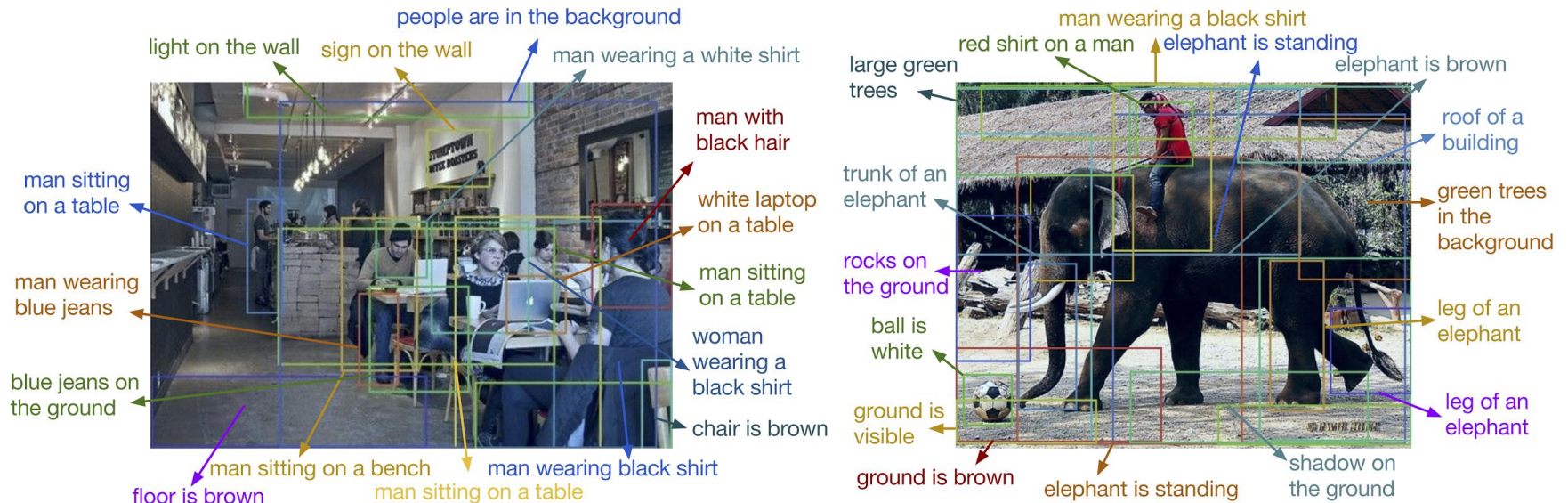


Figure credit: Karpathy and Fei-Fei, "Deep Visual-Semantic Alignments for Generating Image Descriptions", CVPR 2015

Dense Image Captioning



Visual Question Answering



What color are her eyes?
What is the mustache made of?



Is this person expecting company?
What is just under the tree?



How many slices of pizza are there?
Is this a vegetarian pizza?



Does it appear to be rainy?
Does this person have 20/20 vision?

Figure credit: Agrawal et al, "VQA: Visual Question Answering", ICCV 2015

Image			
	Q: Who is behind the batter?	Q: What adorns the tops of the post?	Q: How many cameras are in the photo?
	A: Catcher.	A: Gulls.	A: One.
	A: Umpire.	A: An eagle.	A: Two.
Multiple Choices	A: Fans.	A: A crown.	A: Three.
	A: Ball girl.	A: A pretty sign.	A: Four.
	H: Catcher. ✓	H: Gulls. ✓	H: Three. ✗
	M: Umpire. ✗	M: Gulls. ✓	M: One. ✓
w/ Image w/o Image	H: Catcher. ✓	H: Gulls. ✓	H: One. ✓
	M: Catcher. ✓	M: A crown. ✗	M: One. ✓
			
	Q: Why is there rope?	Q: What kind of stuffed animal is shown?	Q: What animal is being petted?
	A: To tie up the boats.	A: Teddy Bear.	A: A sheep.
	A: To tie up horses.	A: Monkey.	A: Goat.
	A: To hang people.	A: Tiger.	A: Alpaca.
	A: To hit tether balls.	A: Bunny rabbit.	A: Pig.
	H: To hit tether balls. ✗	H: Monkey. ✗	H: A sheep. ✓
	M: To hang people. ✗	M: Teddy Bear. ✓	M: A sheep. ✓
	H: To tie up the boats. ✓	H: Teddy Bear. ✓	H: Goat. ✗
	M: To hang people. ✗	M: Teddy Bear. ✓	M: A sheep. ✓

Figure credit: Zhu et al, "Visual7W: Grounded Question Answering in Images", CVPR 2016

Image Super-Resolution



Figure credit: Ledig et al, "Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network", arXiv 2016

Generating Art

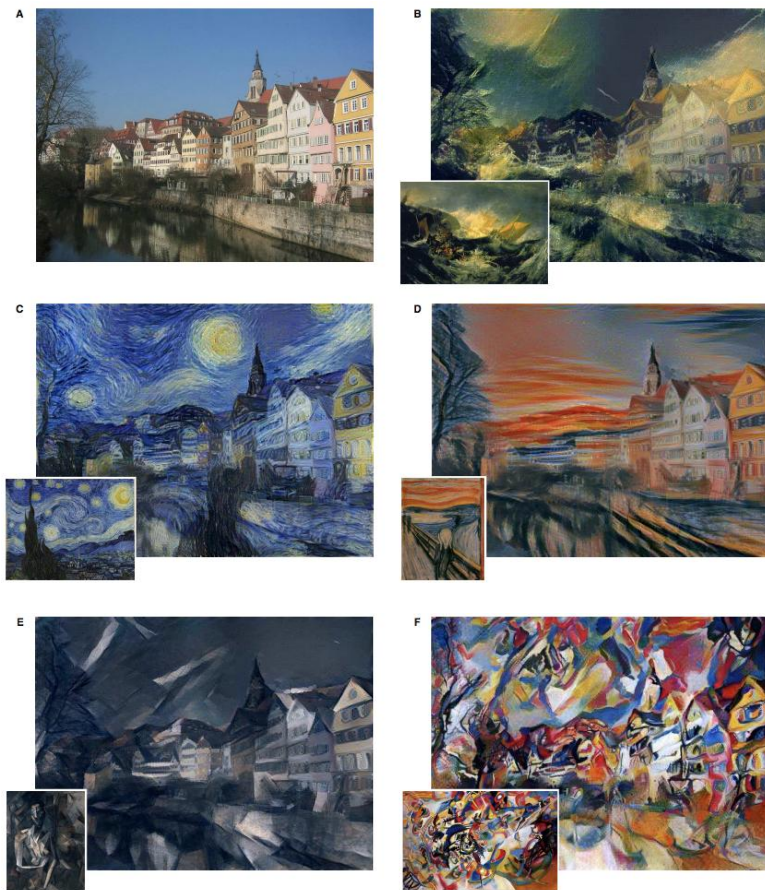


Figure credit: Gatys, Eckert, and Bethge, "Image Style Transfer using Convolutional Neural Networks", CVPR 2016

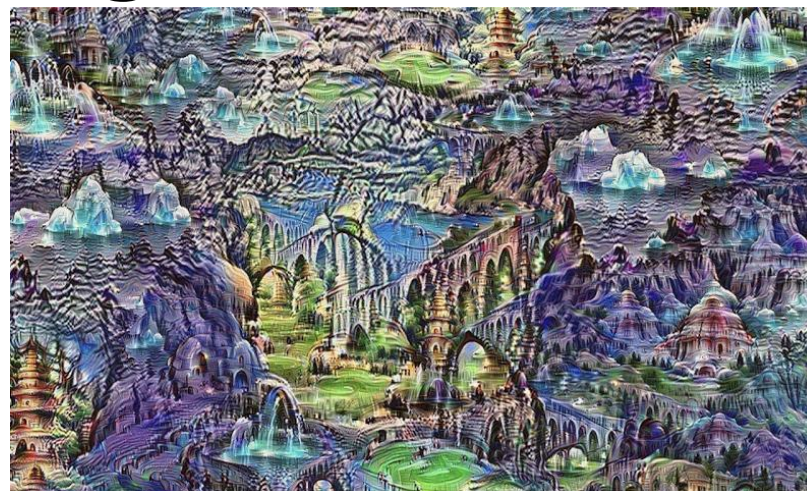


Figure credit: Mordvintsev, Olah, and Tyka, "Inceptionism: Going Deeper into Neural Networks", <https://research.googleblog.com/2015/06/inceptionism-going-deeper-into-neural.html>

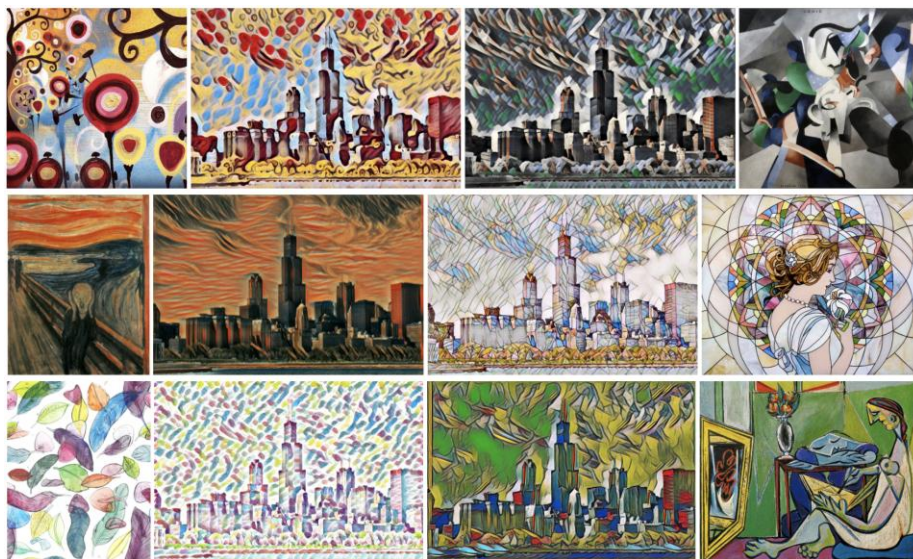
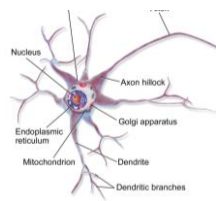
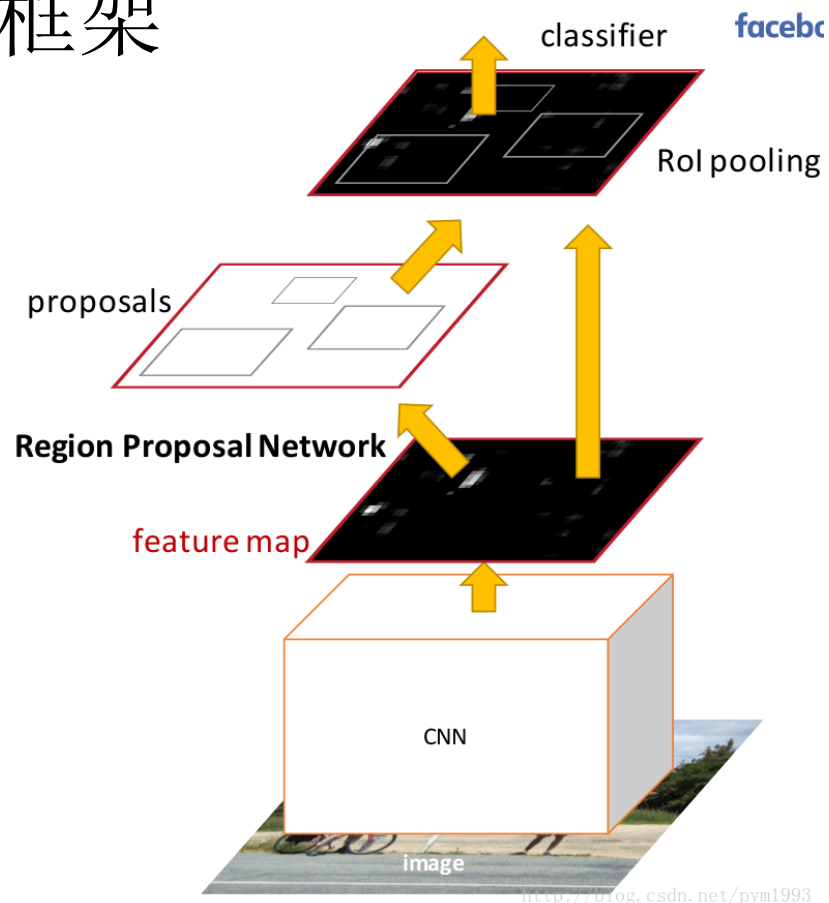


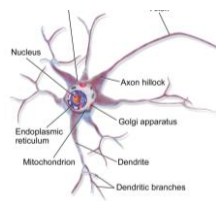
Figure credit: Johnson, Alahi, and Fei-Fei: "Perceptual Losses for Real-Time Style Transfer and Super-Resolution", ECCV 2016, <https://github.com/jcjohnson/fast-neural-style>



目标检测

- 目标检测深度学习框架
 - 候选区+识别模型
 - RCNN
 - Fast RCNN
 - Faster RCNN

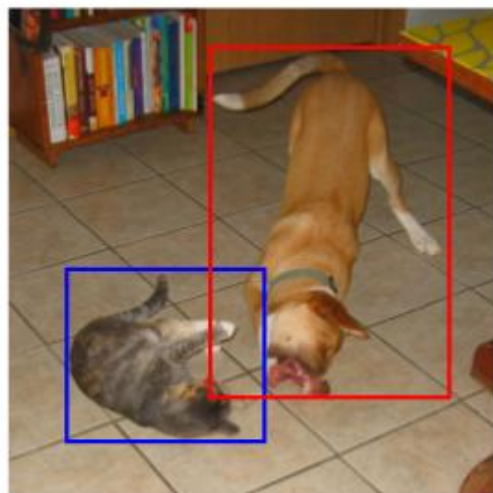




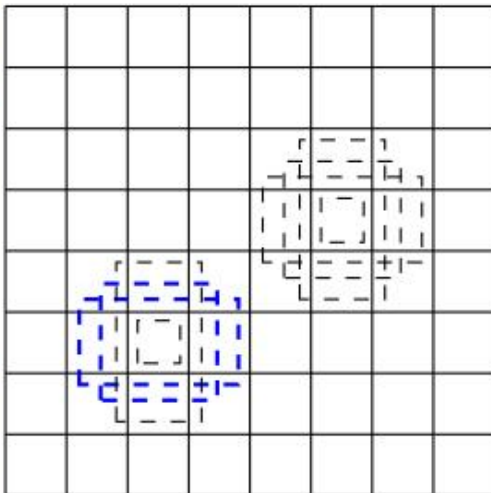
目标检测

- 目标检测深度学习框架
 - 直接回归目标位置
 - SSD
 - YOLO

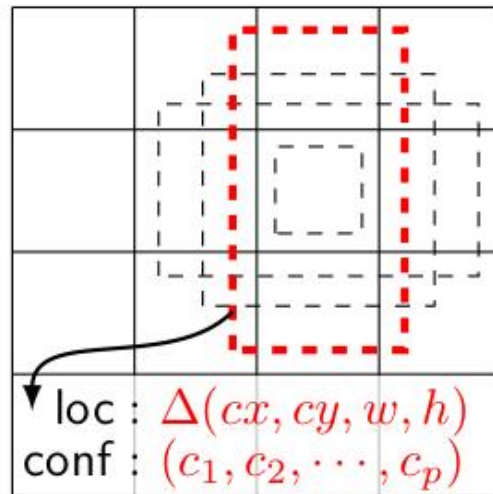
SSD: Single Shot MultiBox Detector



(a) Image with GT boxes

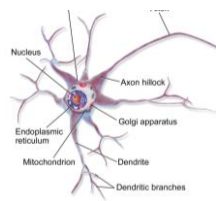


(b) 8×8 feature map



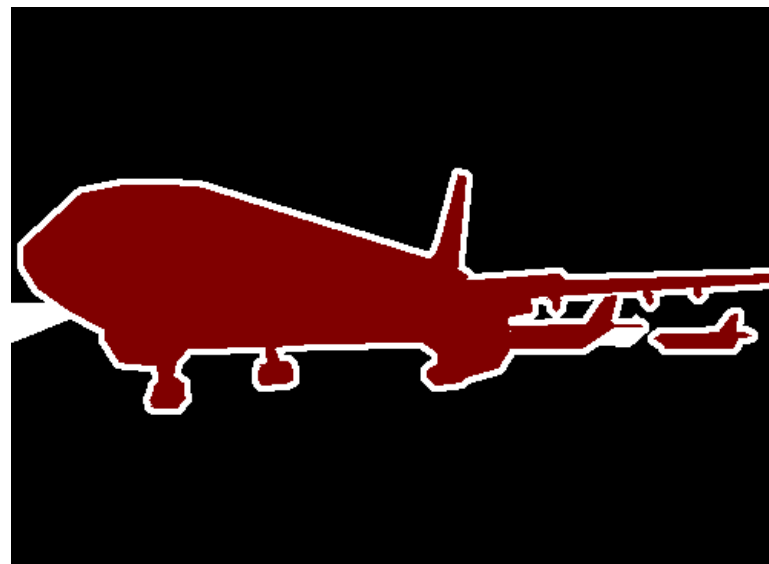
(c) 4×4 feature map

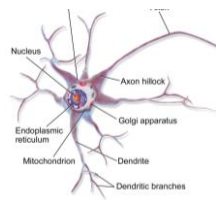
<http://blog.csdn.net/pym1993>



像素级预测

- 语义分割
- 深度估计
- 显著性检测
-





像素级预测

convolution

fully connected



227×227



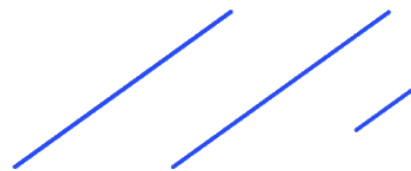
55×55



27×27



13×13



“tabby cat”

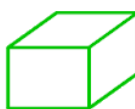
convolution



$H \times W$



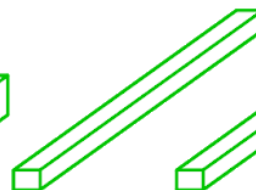
$H/4 \times W/4$



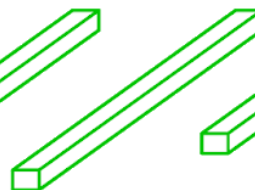
$H/8 \times W/8$



$H/16 \times W/16$

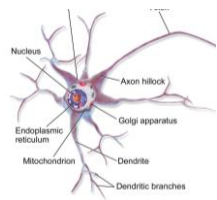


$H/32 \times W/32$

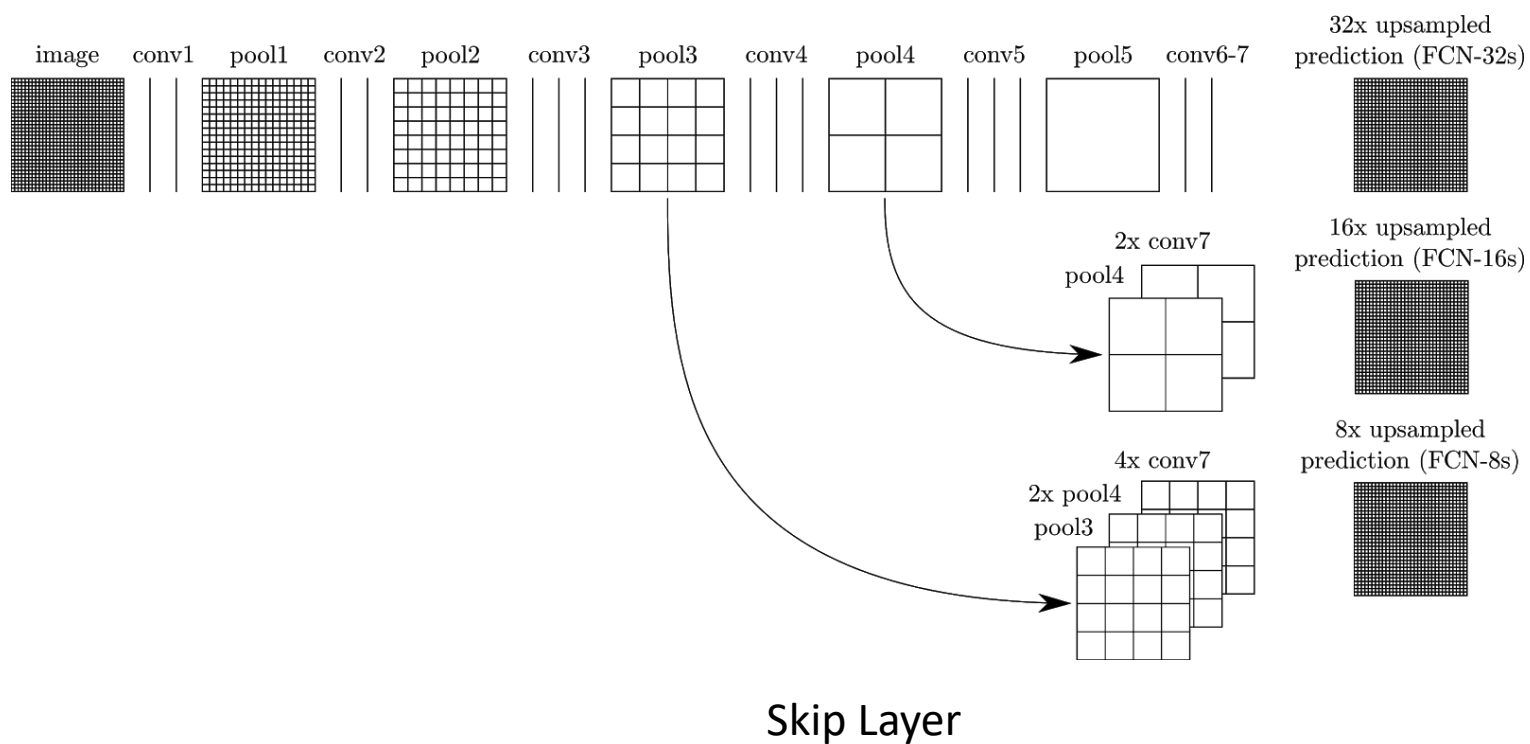


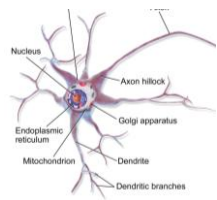
$H \times W$

- Pixel in, pixel out



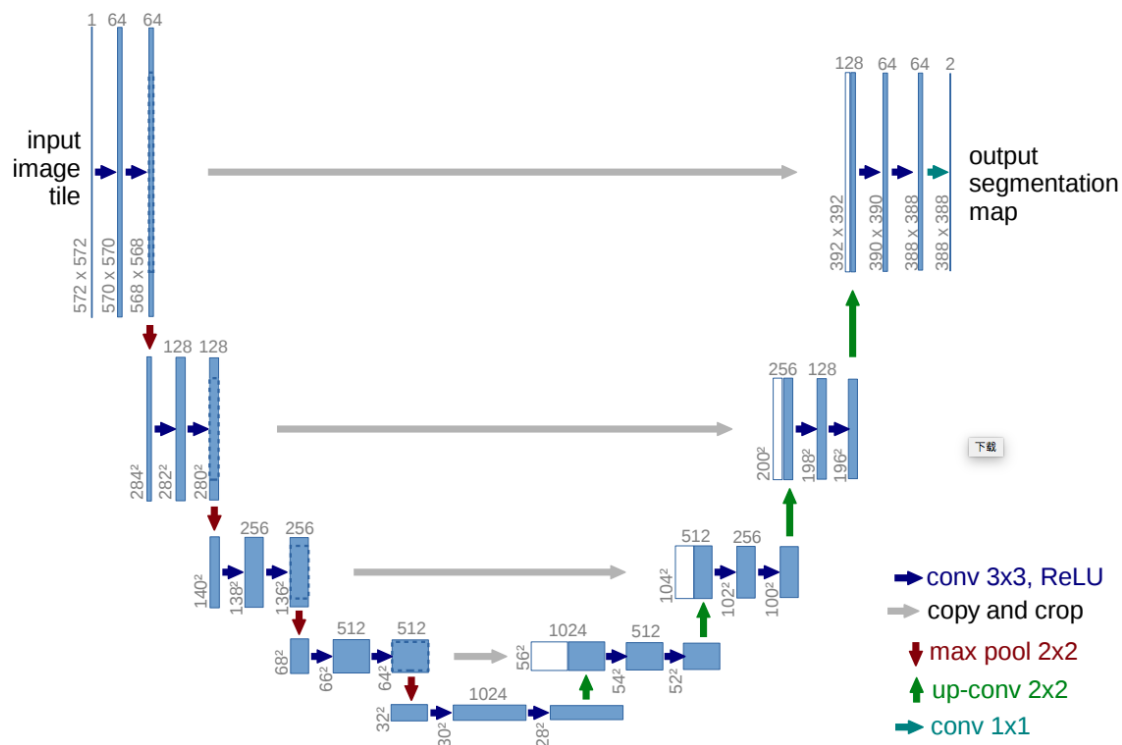
像素级预测

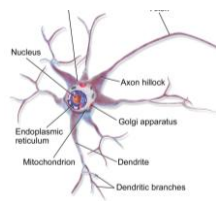




像素级预测

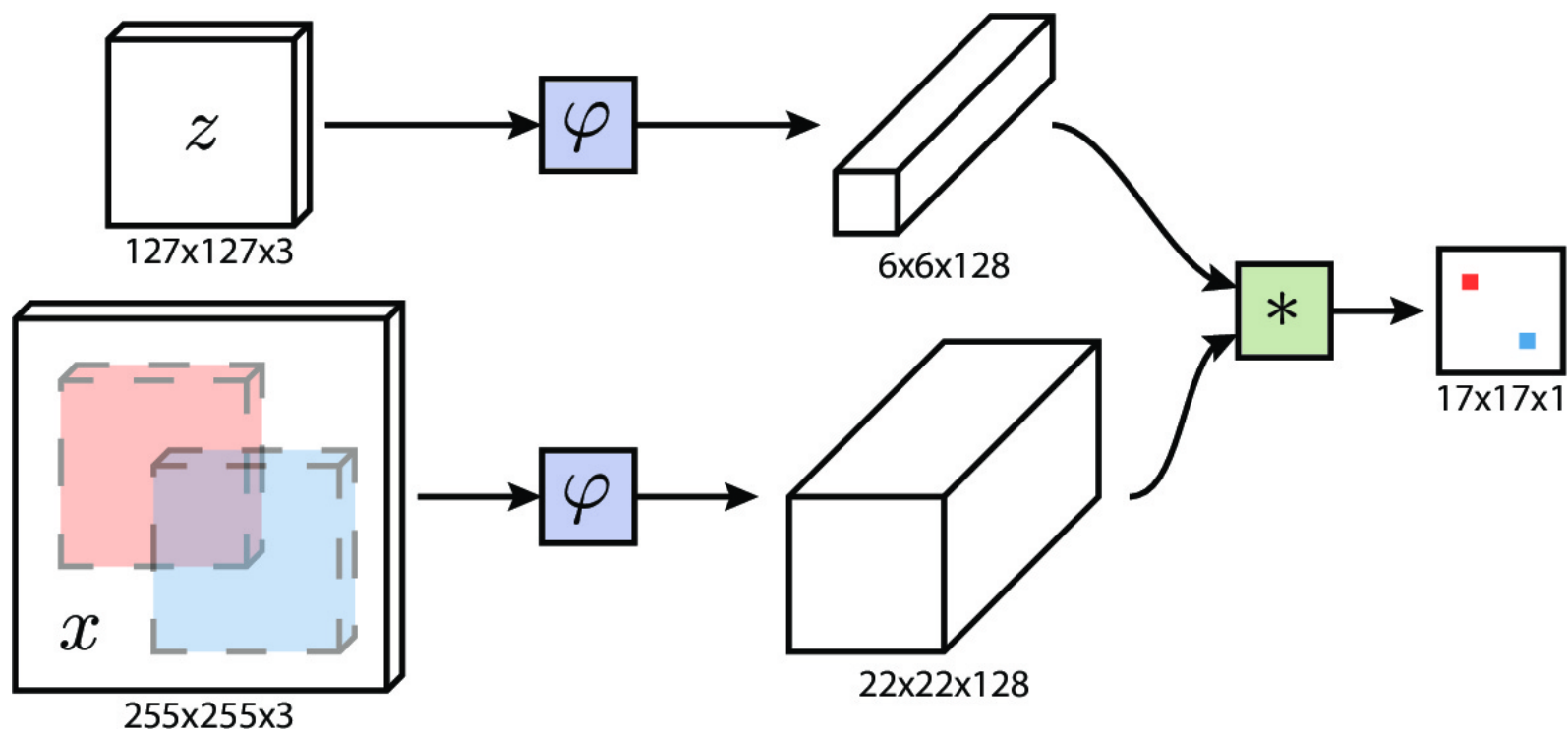
- U-net

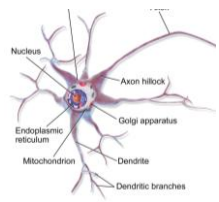




目标跟踪

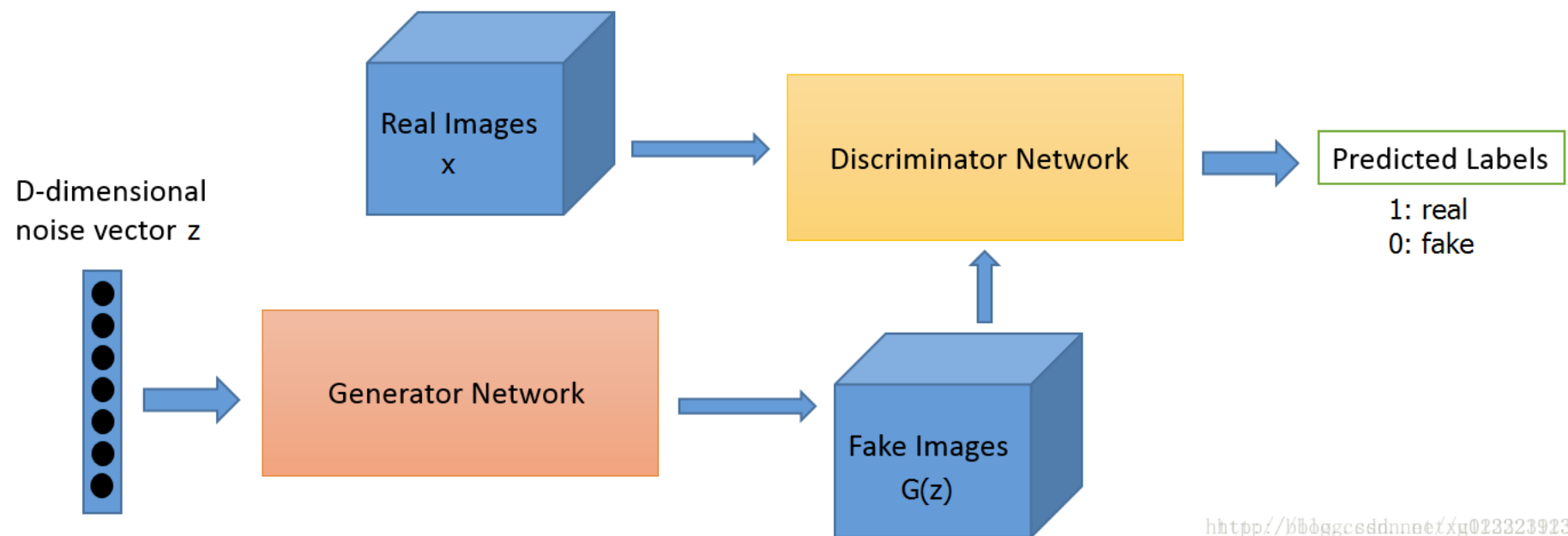
- SiameseFC

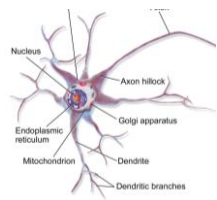




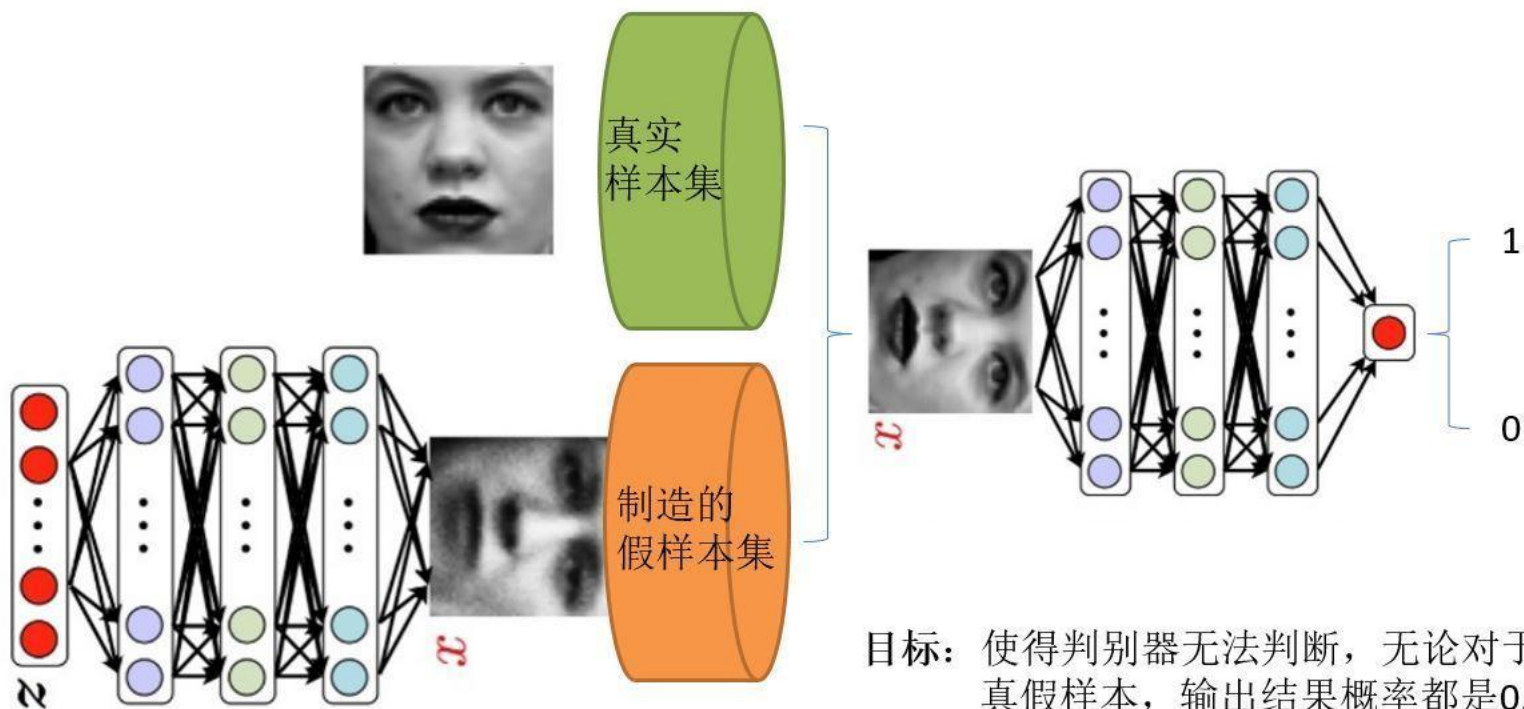
图像生成

- Generative Adversarial Nets (GAN)

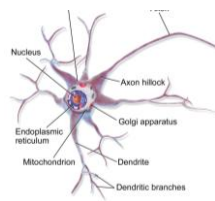




图像生成



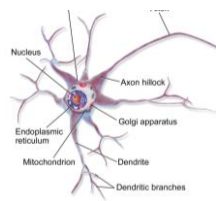
<http://blog.bsdg.net/xdxg423827483>



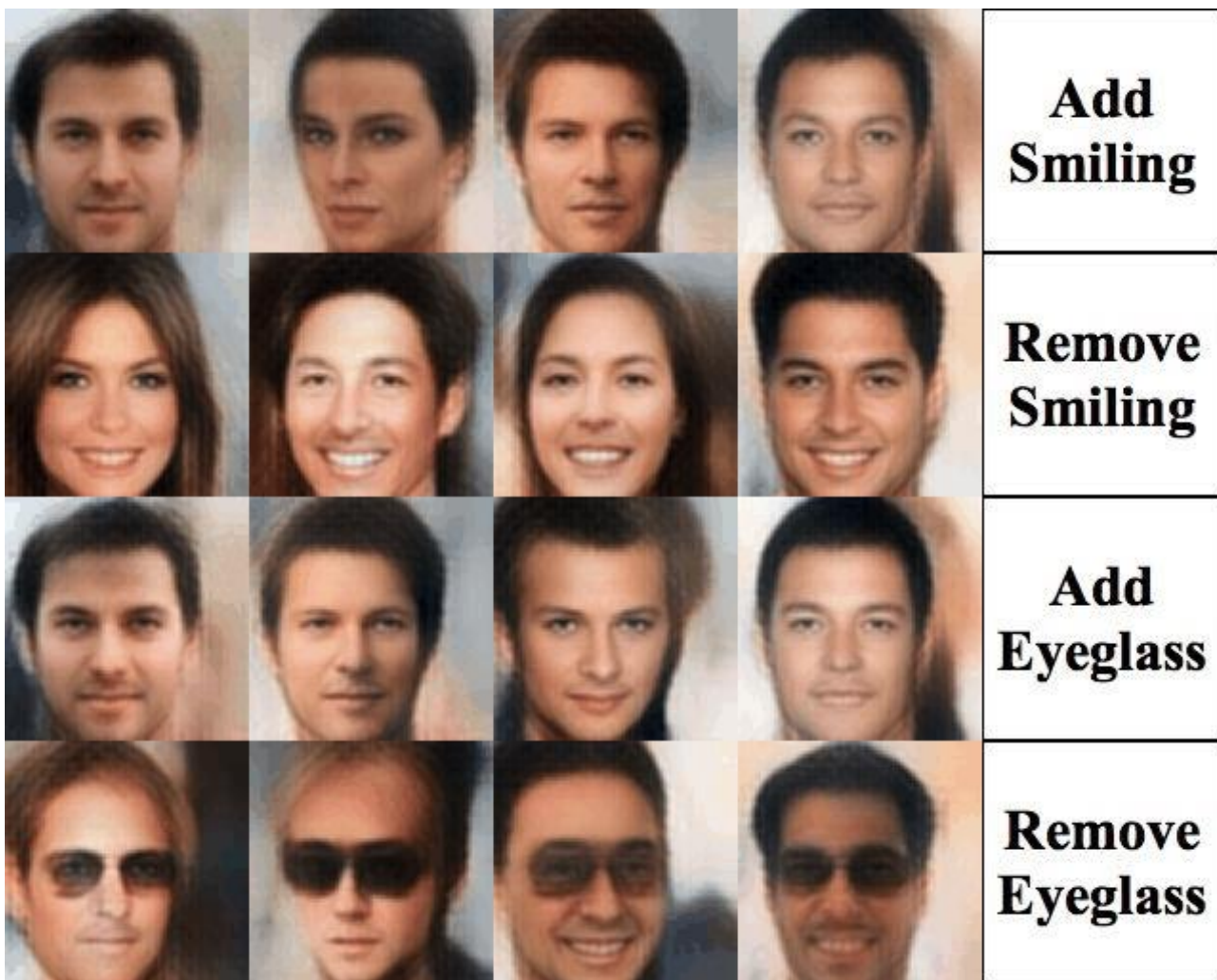
图像生成

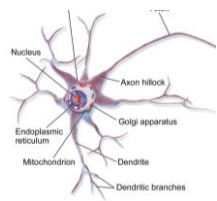
推斷的新方法
字型的進行
東亞語言字
成對抗網路
種類利用條件
字符到字符一

<http://blog.csdn.net/xg123321123>



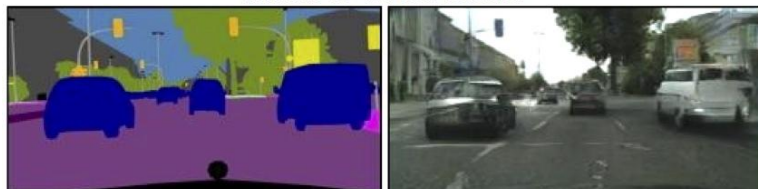
图像生成





图像生成

Labels to Street Scene



input

output

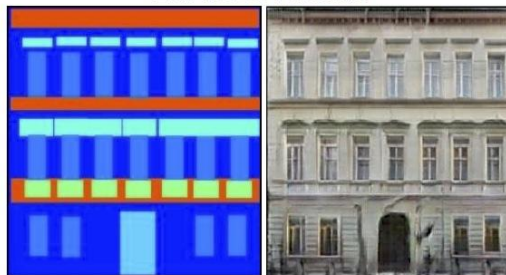
Aerial to Map



input

output

Labels to Facade



input

output

Day to Night



input

output

BW to Color



input

output

Edges to Photo

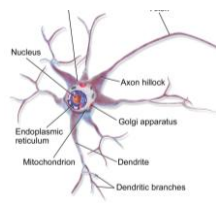


input

<http://blog.cs.123321123> output

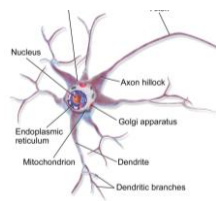


思考



深度学习的局限性与展望

- 缺乏基础理论
- 可解释性差（黑盒子）
- 推理、结构化预测
记忆
经验知识
- 小样本
- 计算复杂度高
- 能用就行
- 暂时不管
- 模拟人脑
- 数据增广、生成
- 模型加速，增强硬件



深度学习的局限性与展望

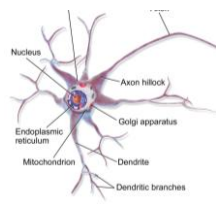
- 深度学习是模拟人脑的处理方式吗？



熊猫
57.7%

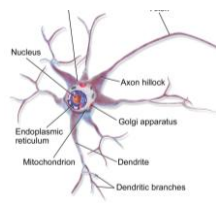


长臂猿
99.3%



深度学习的局限性与展望

- 深度学习是模拟人脑的处理方式吗？
 - 神经元连接方式
 - 分层逐级抽象
- 与人脑的处理机制不同
 - 人脑的“计算机理”尚不清晰
 - 脑皮层连接更多样、更复杂
 - 学习过程未必需要大量数据



谢谢！